S&DS 1000 Class Summaries

Hypothesis Tests and Confidence Intervals

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Randomization Hypothesis Tests

This guide covers every randomization (simulation-based) hypothesis test from S&DS 1000. Each test follows the same five steps used in class:

1. **State** the null and alternative hypotheses
2. **Calculate** the observed test statistic
3. **Create** a null distribution by simulation
4. **Calculate** the p-value
5. **Make a decision**

The core idea is the same for every test: if the null hypothesis were true, how extreme would our observed result be? We answer this by repeatedly simulating data under the null and seeing where our observed statistic falls.

> Tip
>
> **Key SDS1000 functions used in this guide.** `do_it(n) * { expr }` repeats an expression `n` times and collects the results. `rflip(n, p)` simulates `n` coin flips with probability `p`. `rroll(n, prob)` simulates rolling a die `n` times. `shuffle(x)` randomly permutes a vector. `ptail(obs, null, lower.tail)` computes the p-value from a null distribution.

One Proportion

When to use: One categorical variable with two outcomes; test whether the population proportion equals a specific value . The null distribution is created by simulating coin flips under the assumption that H is true using `rflip()`.

Example (class 12): Doris the dolphin and Buzz the dolphin were tested to see whether Buzz could identify which lever Doris was pointing to. In 16 trials, Buzz got the correct answer 15 times. Is there evidence that Buzz is doing better than random chance?

Step 1: State the hypotheses

$$H_0 : \pi = 0.50 \qquad H_A : \pi > 0.50$$

Step 2: Calculate the observed statistic

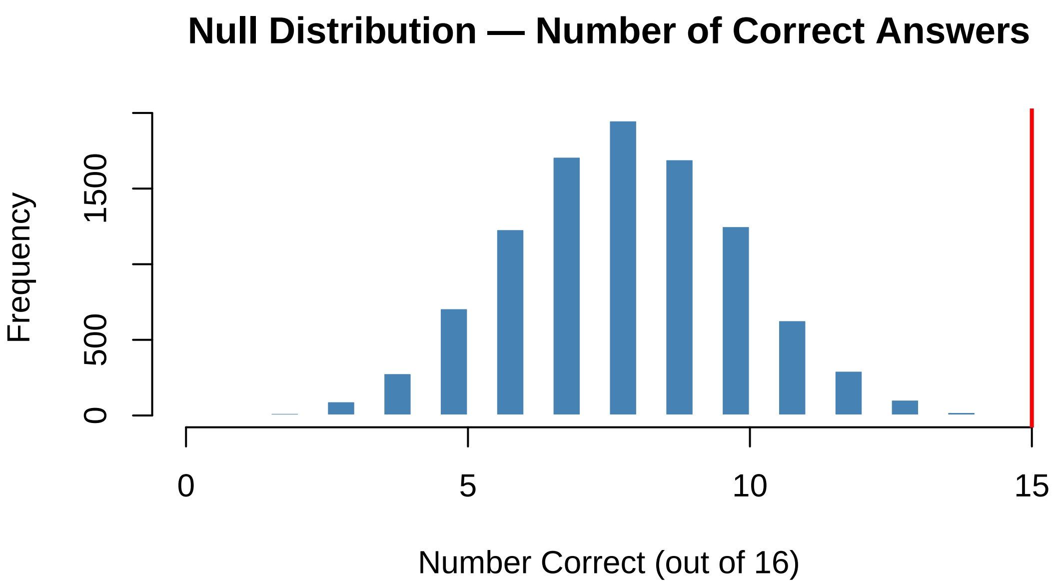
```r
n_trials <- 16
obs_stat <- 15   # number of correct answers
obs_stat
```

```
[1] 15
```

Step 3: Create the null distribution

Under H , each trial is like a fair coin flip. Use `rflip()` to simulate 10,000 experiments:

```
set.seed(5731)

null_dist <- do_it(10000) * {
  rflip(n_trials, 0.50)
}

hist(null_dist, breaks = 50,
     main = "Null Distribution - Number of Correct Answers",
     xlab = "Number Correct (out of 16)",
     col = "steelblue", border = "white")
abline(v = obs_stat, col = "red", lwd = 2)
```



Step 4: Calculate the p-value

```
p_value <- ptail(obs_stat, null_dist, lower.tail = FALSE)
p_value
```

```
[1] 5e-04
```

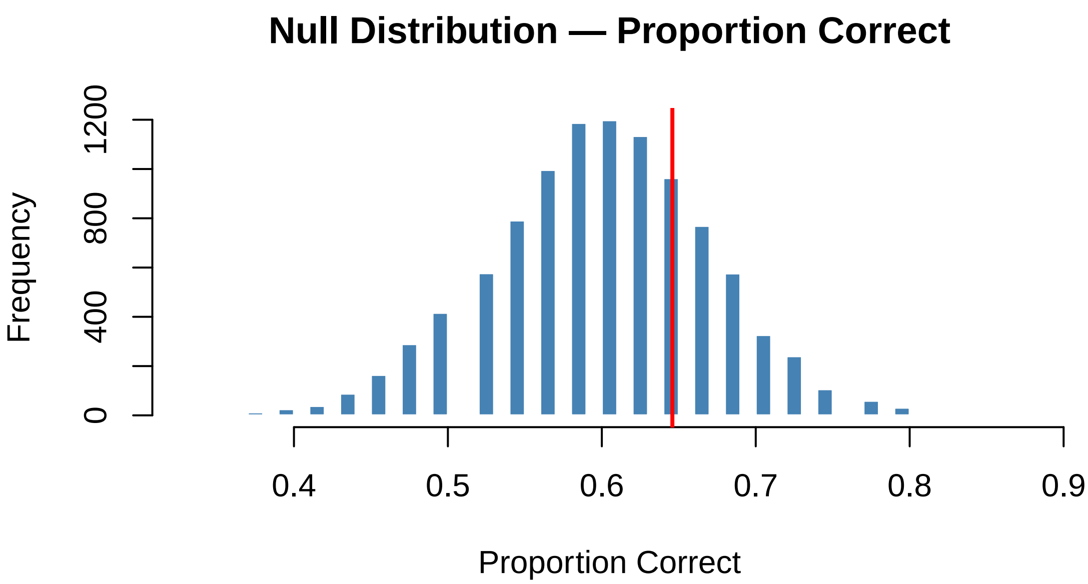
Step 5: Make a decision

Since $p = 0.0005 < 0.05$, we **reject** H and conclude that Buzz appears to be doing better than random chance.

Second example (class 13): A lie detector correctly identified 31 out of 48 participants as lying. Is this evidence the detector is more than 60% accurate?

$$H_0 : \pi = 0.60 \qquad H_A : \pi > 0.60$$

```r
set.seed(9021)

n        <- 48
obs_phat <- 31 / 48

null_dist <- do_it(10000) * {
  rflip(n, 0.60) / n
}

hist(null_dist, breaks = 60,
     main = "Null Distribution - Proportion Correct",
     xlab = "Proportion Correct",
     col = "steelblue", border = "white")
abline(v = obs_phat, col = "red", lwd = 2)
```




```r
p_value <- ptail(obs_phat, null_dist, lower.tail = FALSE)
p_value
```

```
[1] 0.3084
```

Since p = 0.308 > 0.05, we **fail to reject** H . There is not sufficient evidence that the lie detector exceeds 60% accuracy.

Two Proportions

When to use: A binary outcome measured in two independent groups; test whether the two population proportions are equal.

Key idea: Under H (=), both groups share the same true proportion. We estimate this **overall proportion** by pooling the two samples:

$$\hat{p}_{\text{overall}} = \frac{x_1 + x_2}{n_1 + n_2}$$

We then simulate each group independently using this pooled rate with **`rflip()`**, mimicking what we would observe if H were true.

Example: In a clinical trial, 245 patients received a new vaccine and 255 received a placebo. Of the vaccinated group 32 got the flu; of the placebo group 56 got the flu. Is there evidence that the vaccine reduces the flu rate?

Step 1: State the hypotheses

Let group 1 = vaccine, group 2 = placebo.

$$H_0 : \pi_1 = \pi_2 \qquad H_A : \pi_1 < \pi_2$$

Step 2: Calculate the observed statistic

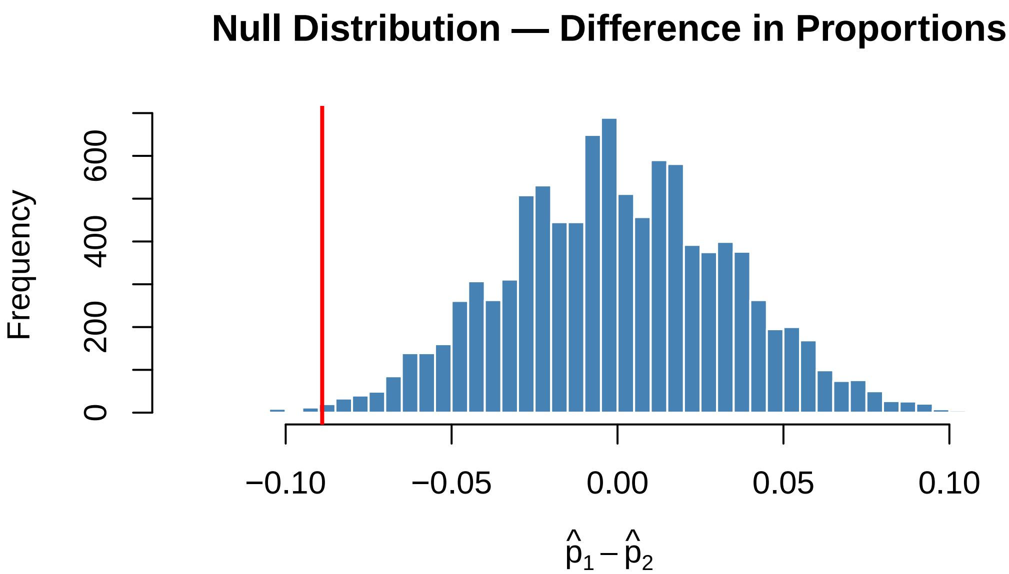
```r
x1 <- 32;  n1 <- 245   # vaccinated
x2 <- 56;  n2 <- 255   # placebo

obs_phat1 <- x1 / n1
obs_phat2 <- x2 / n2
obs_stat  <- obs_phat1 - obs_phat2
obs_stat
```

```
[1] -0.0889956
```

Step 3: Create the null distribution

```r
set.seed(4412)

overall_prop <- (x1 + x2) / (n1 + n2)

null_dist <- do_it(10000) * {
  sim_phat1 <- rflip(n1, overall_prop) / n1
  sim_phat2 <- rflip(n2, overall_prop) / n2
  sim_phat1 - sim_phat2
```

```
}

hist(null_dist, breaks = 80,
     main = "Null Distribution - Difference in Proportions",
     xlab = expression(hat(p)[1] - hat(p)[2]),
     col = "steelblue", border = "white")
abline(v = obs_stat, col = "red", lwd = 2)
```



Step 4: Calculate the p-value

```
p_value <- ptail(obs_stat, null_dist, lower.tail = TRUE)
p_value
```

```
[1] 0.0034
```

Step 5: Make a decision

Since $p = 0.0034 < 0.05$, we **reject** H and conclude that the vaccine is associated with a significantly lower flu rate.

More Than Two Proportions

When to use: One categorical variable with three or more outcomes; test whether the population proportions match a set of expected values. The observed statistic is the chi-squared statistic; the null distribution is created by simulating random samples using `rroll()`.

Example (class 23): A Yale professor examined the birth months of 198 randomly selected Yale students. Are students equally likely to be born in any month?

Step 1: State the hypotheses

$$H_0 : \pi_{Jan} = \pi_{Feb} = \cdots = \pi_{Dec} = \frac{1}{12}$$
$$H_A : \text{at least one monthly proportion differs from } \frac{1}{12}$$

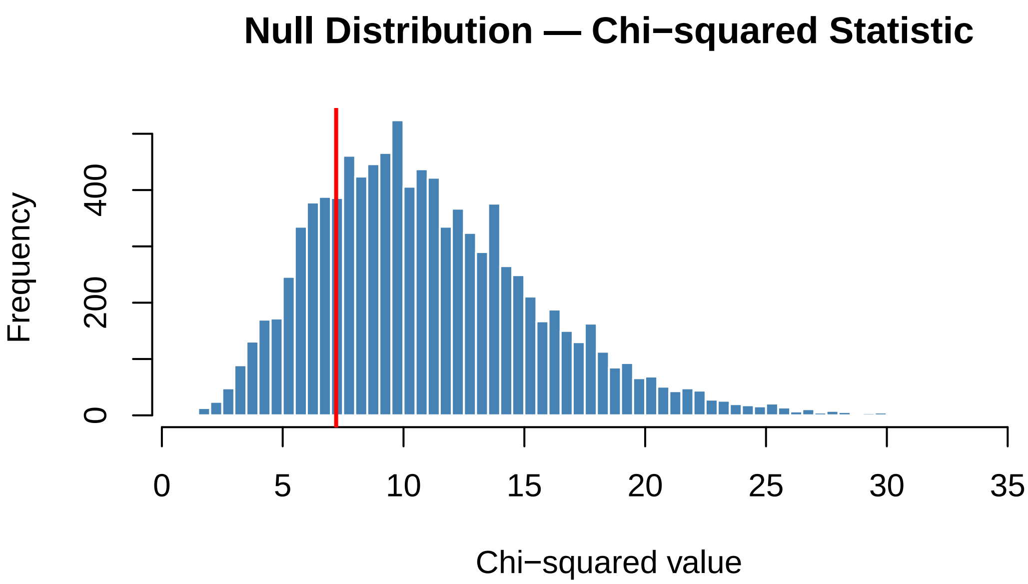
Step 2: Calculate the observed statistic

```
observed_counts <- c(14, 11, 21, 17, 15, 13, 19, 16, 18, 22, 14, 18)
names(observed_counts) <- month.abb

expected_counts <- rep(sum(observed_counts) / 12, 12)

obs_stat <- sum((observed_counts - expected_counts)^2 / expected_counts)
obs_stat
```

```
[1] 7.212121
```

Step 3: Create the null distribution

```
set.seed(3341)

n_students <- sum(observed_counts)

null_dist <- do_it(10000) * {
  sim_counts <- rroll(n_students, prob = rep(1/12, 12))
  sum((sim_counts - expected_counts)^2 / expected_counts)
}

hist(null_dist, breaks = 60,
     main = "Null Distribution - Chi-squared Statistic",
     xlab = "Chi-squared value",
     col = "steelblue", border = "white")
abline(v = obs_stat, col = "red", lwd = 2)
```



Step 4: Calculate the p-value

```
p_value <- ptail(obs_stat, null_dist, lower.tail = FALSE)
p_value
```

```
[1] 0.7917
```

Step 5: Make a decision

Since p = 0.792 > 0.05, we **fail to reject** H . There is no evidence that Yale students' birth months deviate from a uniform distribution.

One Mean

When to use: A single quantitative variable; test whether the population mean equals a specific value . We use a **bootstrap-based procedure**: shift the data so its mean equals , then resample from that shifted data.

Key idea:

```
data_H0 <- my_data - mean(my_data) + mu_0
```

Example: It is widely assumed that the average human body temperature is 98.6°F. A sample of 50 body temperatures had a mean of 98.25°F. Is there evidence that the true mean differs from 98.6°F?

Step 1: State the hypotheses

$$H_0 : \mu = 98.6 \qquad H_A : \mu \neq 98.6$$

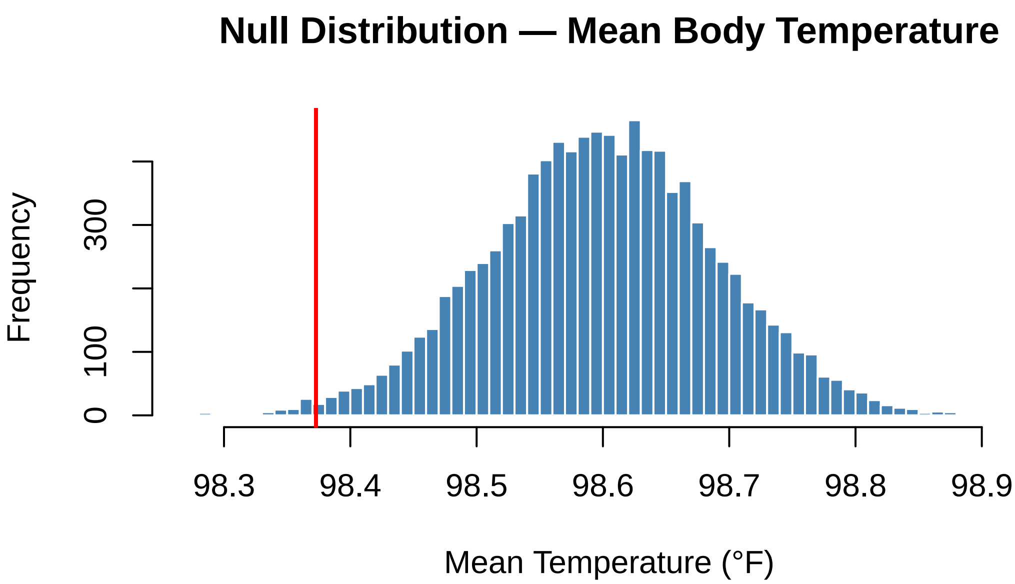
Step 2: Calculate the observed statistic

```
set.seed(9910)
body_temps <- rnorm(50, mean = 98.25, sd = 0.73)

obs_mean <- mean(body_temps)
obs_mean
```

```
[1] 98.37283
```

Step 3: Create the null distribution

```
set.seed(3345)

mu_0    <- 98.6
data_H0 <- body_temps - mean(body_temps) + mu_0
n       <- length(body_temps)

null_dist <- do_it(10000) * {
  mean(sample(data_H0, n, replace = TRUE))
}

hist(null_dist, breaks = 80,
     main = "Null Distribution - Mean Body Temperature",
     xlab = "Mean Temperature (°F)",
     col = "steelblue", border = "white")
abline(v = obs_mean, col = "red", lwd = 2)
```



Step 4: Calculate the p-value

Two-sided test — count values at least as far from as the observed mean:

```
margin <- abs(obs_mean - mu_0)

p_value <- ptail(mu_0 - margin, null_dist, lower.tail = TRUE) +
           ptail(mu_0 + margin, null_dist, lower.tail = FALSE)
p_value
```

```
[1] 0.0114
```

Step 5: Make a decision

Since $p = 0.0114 < 0.05$, we **reject** H and conclude that there is evidence the true mean body temperature differs from 98.6°F.

Two Means

When to use: A quantitative variable measured in two independent groups; test whether the two group means are equal. The null distribution is created by using `shuffle()` to randomly reassign group labels.

Example (class 17): A randomized controlled trial gave calcium supplements to 10 men (treatment) and a placebo to 11 men (control). The outcome is the decrease in blood pressure. Is there evidence that calcium produces a greater decrease?

Step 1: State the hypotheses

$$H_0 : \mu_{\text{treat}} = \mu_{\text{control}} \qquad H_A : \mu_{\text{treat}} > \mu_{\text{control}}$$

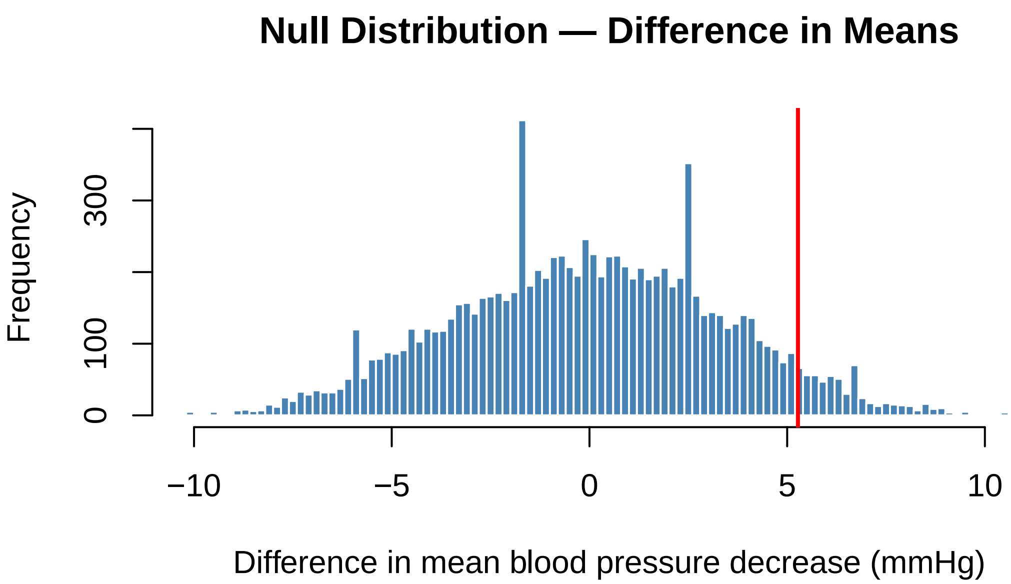
Step 2: Calculate the observed statistic

```
treat   <- c( 7, -4, 18, 17, -3, -5,  1, 10, 11, -2)
control <- c(-1, 12, -1, -3,  3, -5,  5,  2, -11, -1, -3)

obs_stat <- mean(treat) - mean(control)
obs_stat
```

```
[1] 5.272727
```

Step 3: Create the null distribution

```
set.seed(6174)

combined_data <- c(treat, control)

null_dist <- do_it(10000) * {
  shuff         <- shuffle(combined_data)
  shuff_treat   <- shuff[1:10]
  shuff_control <- shuff[11:21]
  mean(shuff_treat) - mean(shuff_control)
}

hist(null_dist, breaks = 100,
     main = "Null Distribution - Difference in Means",
     xlab = "Difference in mean blood pressure decrease (mmHg)",
     col = "steelblue", border = "white")
abline(v = obs_stat, col = "red", lwd = 2)
```



Step 4: Calculate the p-value

```
p_value <- ptail(obs_stat, null_dist, lower.tail = FALSE)
p_value
```

```
[1] 0.0605
```

Step 5: Make a decision

Since p = 0.06 is just above 0.05, we **fail to reject** H at the = 0.05 level. The evidence for a calcium effect is suggestive but not conclusive.

More Than Two Means

When to use: A quantitative variable measured across three or more independent groups; test whether all group means are equal. The observed statistic is the **mean absolute deviation (MAD)** of group means. The null distribution is created by using `shuffle()` on the group labels.

Example (class 17/18): Hope College students were timed completing a Sudoku-like puzzle, grouped into four majors. Is there a difference in mean completion time?

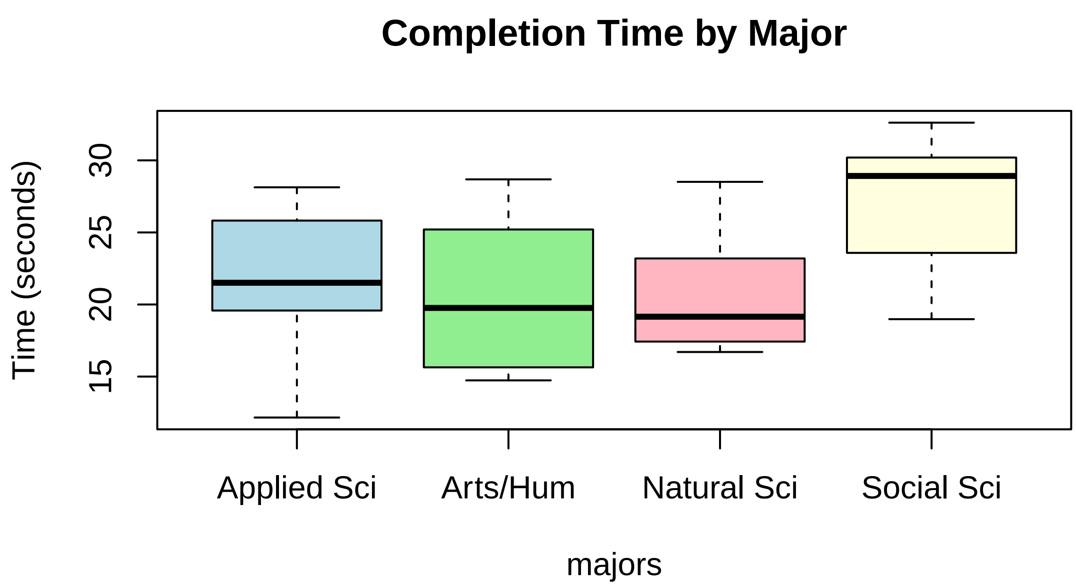
Step 1: State the hypotheses

$$H_0 : \mu_{as} = \mu_{ns} = \mu_{ss} = \mu_{ah}$$

$$H_A : \text{at least one group mean differs}$$

Step 2: Calculate the observed statistic

```
set.seed(4455)
completion_times <- c(rnorm(10, mean = 22, sd = 5),
                      rnorm(10, mean = 20, sd = 5),
                      rnorm(10, mean = 26, sd = 5),
                      rnorm(10, mean = 21, sd = 5))
majors <- rep(c("Applied Sci", "Natural Sci", "Social Sci", "Arts/Hum"),
              each = 10)

boxplot(completion_times ~ majors,
        main = "Completion Time by Major",
        ylab = "Time (seconds)",
        col  = c("lightblue", "lightgreen", "lightpink", "lightyellow"))
```



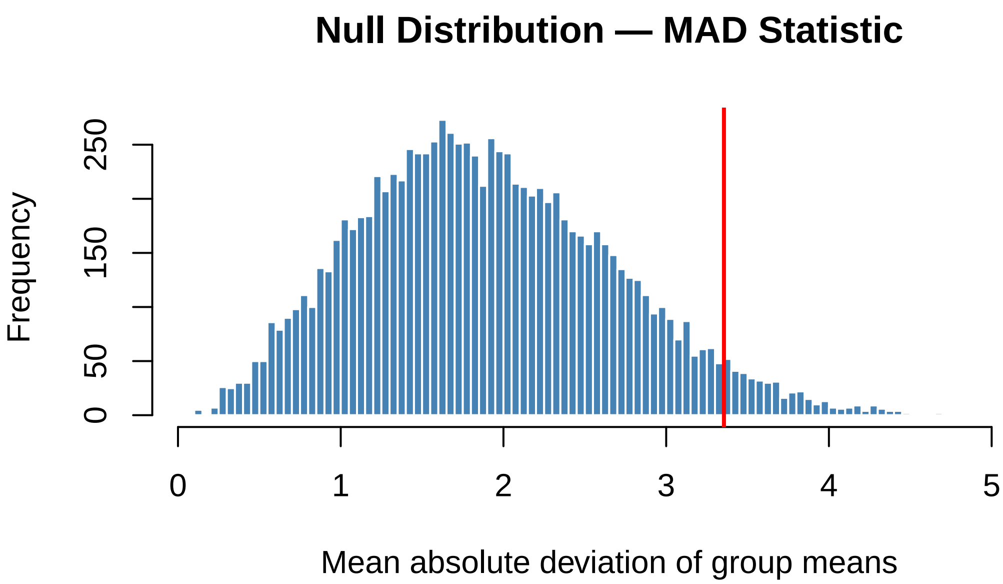

```
obs_stat <- get_MAD_stat(completion_times, majors)
obs_stat
```

```
[1] 3.354879
```

Step 3: Create the null distribution

```
set.seed(8812)

null_dist <- do_it(10000) * {
  get_MAD_stat(completion_times, shuffle(majors))
}
```

```r
hist(null_dist, breaks = 100,
     main = "Null Distribution - MAD Statistic",
     xlab = "Mean absolute deviation of group means",
     col = "steelblue", border = "white")
abline(v = obs_stat, col = "red", lwd = 2)
```



Step 4: Calculate the p-value

```r
p_value <- ptail(obs_stat, null_dist, lower.tail = FALSE)
p_value
```

```
[1] 0.0416
```

Step 5: Make a decision

Since p = 0.042, we **reject** H at the   = 0.05 level.

> **Tip**
>
> **What is the MAD statistic?** `get_MAD_stat()` computes the mean of all pairwise absolute differences between group means:
>
> $$MAD = \frac{1}{\binom{k}{2}} \sum_{i<j} |\bar{x}_i - \bar{x}_j|$$
>
> It is sensitive to any difference between group means, making it a natural choice for a general alternative hypothesis. Any statistic can be used — for example, `get_F_stat()` would give the ANOVA F-statistic.

Correlation

When to use: Two quantitative variables; test whether the population correlation equals zero. The null distribution is created by using `shuffle()` on one variable — this breaks any real relationship while preserving the marginal distributions.

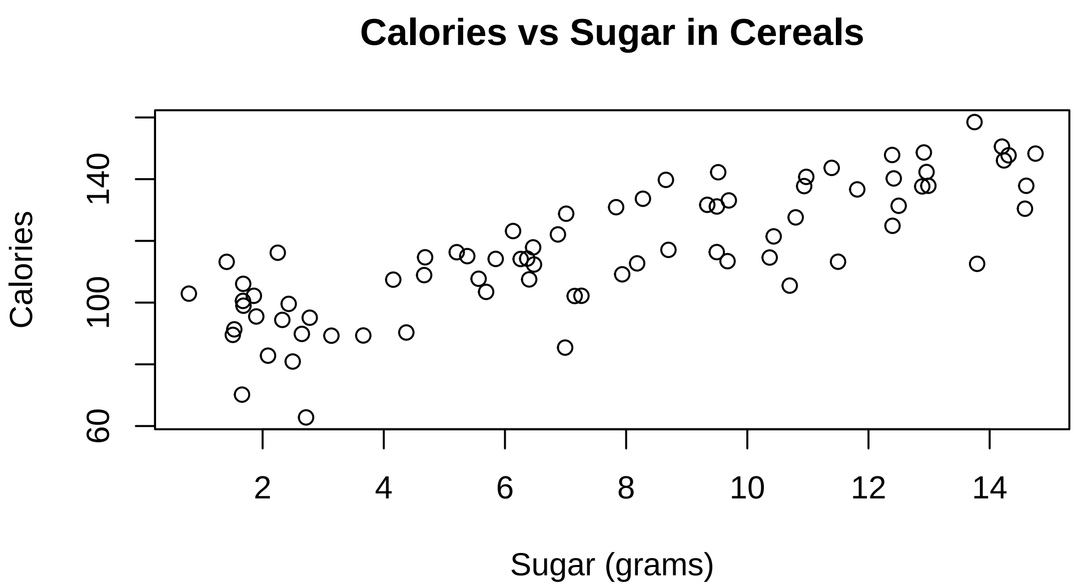
Example (class 18): A dataset of 77 breakfast cereals recorded sugar content and calorie content. Is there evidence of a positive correlation?

Step 1: State the hypotheses

$$H_0 : \rho = 0 \qquad H_A : \rho > 0$$

Step 2: Calculate the observed statistic

```
set.seed(1123)
n        <- 77
sugar    <- runif(n, 0, 15)
calories <- 90 + 3.5 * sugar + rnorm(n, sd = 12)

plot(sugar, calories,
     main = "Calories vs Sugar in Cereals",
     xlab = "Sugar (grams)", ylab = "Calories")
```

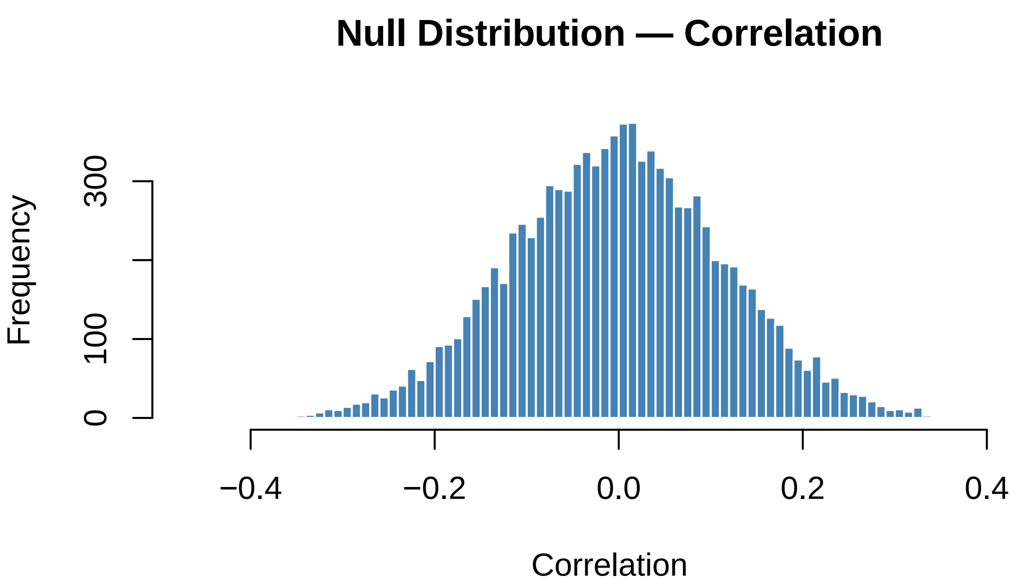



```
obs_stat <- cor(sugar, calories)
obs_stat
```

```
[1] 0.8149169
```

Step 3: Create the null distribution

```r
set.seed(5671)

null_dist <- do_it(10000) * {
  cor(shuffle(sugar), calories)
}

hist(null_dist, breaks = 80,
     main = "Null Distribution - Correlation",
     xlab = "Correlation",
     col = "steelblue", border = "white")
abline(v = obs_stat, col = "red", lwd = 2)
```



Step 4: Calculate the p-value

```r
p_value <- ptail(obs_stat, null_dist, lower.tail = FALSE)
p_value
```

```
[1] 0
```

Step 5: Make a decision

Since p 0, we **reject** H and conclude that there is evidence of a positive correlation between sugar and calorie content in cereals.

Simple Linear Regression

When to use: Two quantitative variables where you want to test whether the slope of a linear regression model is significantly different from zero. The null distribution is created by using `shuffle()` on the response variable.

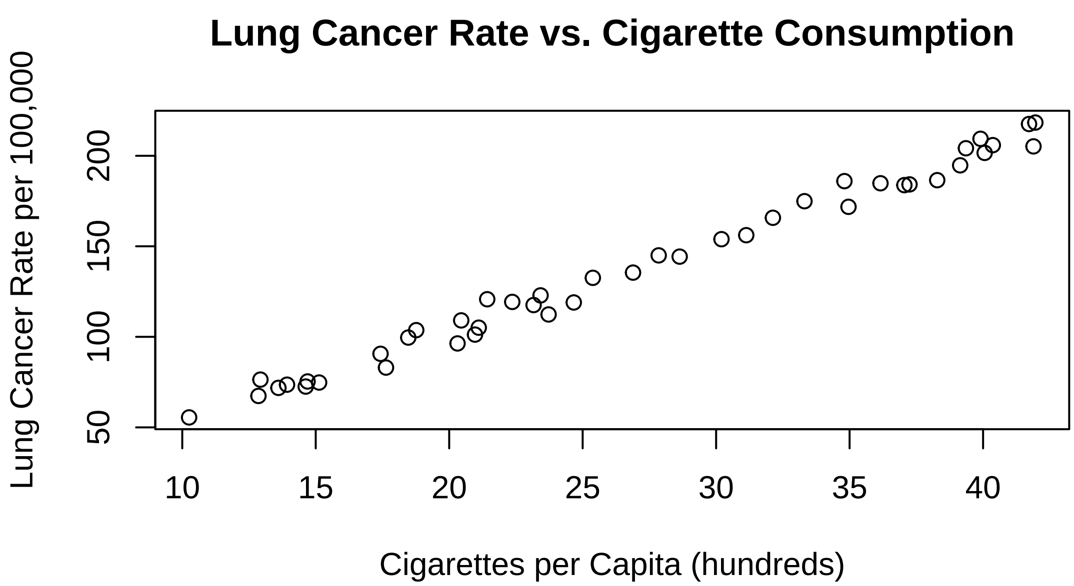
Example (class 25): Data on cigarette consumption per capita and lung cancer rates across U.S. states. Is there evidence of a positive slope?

Step 1: State the hypotheses

$$H_0 : \beta_1 = 0 \qquad H_A : \beta_1 > 0$$

Step 2: Calculate the observed statistic

```
set.seed(2947)
cigs_per_capita <- runif(44, min = 10, max = 45)
cancer_rate     <- 2 + 0.005 * cigs_per_capita * 1000 + rnorm(44, sd = 5)

plot(cigs_per_capita, cancer_rate,
     xlab = "Cigarettes per Capita (hundreds)",
     ylab = "Lung Cancer Rate per 100,000",
     main = "Lung Cancer Rate vs. Cigarette Consumption")
```

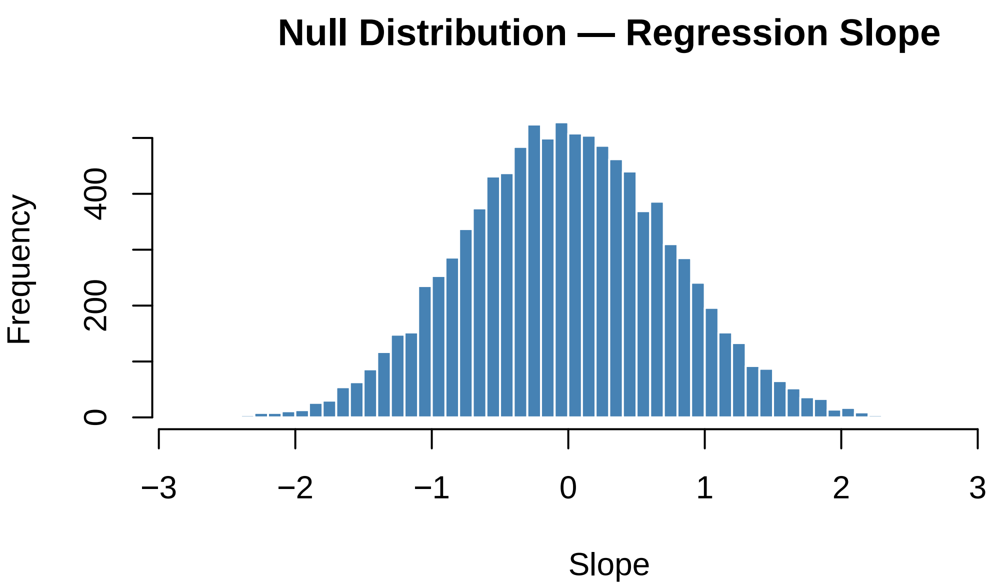



```
lung_lm   <- lm(cancer_rate ~ cigs_per_capita)
obs_slope <- coef(lung_lm)["cigs_per_capita"]
obs_slope
```

```
cigs_per_capita
       4.997922
```

Step 3: Create the null distribution

```
set.seed(3318)

null_dist <- do_it(10000) * {
  null_lm <- lm(shuffle(cancer_rate) ~ cigs_per_capita)
  coef(null_lm)["cigs_per_capita"]
}

hist(null_dist, breaks = 80,
     main = "Null Distribution - Regression Slope",
     xlab = "Slope",
     col = "steelblue", border = "white")
abline(v = obs_slope, col = "red", lwd = 2)
```



Step 4: Calculate the p-value

```
p_value <- ptail(obs_slope, null_dist, lower.tail = FALSE)
p_value
```

```
[1] 0
```

Step 5: Make a decision

Since p 0, we **reject** H and conclude there is evidence of a positive relationship between cigarette consumption and lung cancer rate.

Quick Reference

Test	Observed statistic	How to simulate the null	p-value
One proportion	Count or proportion	`do_it(n) * { rflip(n, pi_0) }`	`ptail(obs, null, lower.tail = ...)`
Two proportions	$\hat{p}_1 - \hat{p}_2$	`do_it(n) * { rflip(n1,p_pool)/n1 - rflip(n2,p_pool)/n2 }`	`ptail(obs, null, lower.tail = ...)`
Multiple proportions	$\sum(O-E)^2/E$	`do_it(n) * { rroll(n, expected_p) }`	`ptail(obs, null, lower.tail = FALSE)`
One mean	$\bar{x}$	Shift data to , then resample	two-sided sum of tails
Two means	$\bar{x}_1 - \bar{x}_2$	`do_it(n) * { shuff <- shuffle(combined); ... }`	`ptail(obs, null, lower.tail = ...)`
More than two means	MAD: `get_MAD_stat()`	`do_it(n) * { get_MAD_stat(y, shuffle(groups)) }`	`ptail(obs, null, lower.tail = FALSE)`
Correlation	$r =$ `cor(x, y)`	`do_it(n) * { cor(shuffle(x), y) }`	`ptail(obs, null, lower.tail = ...)`
Regression slope	$\hat{\beta}_1$ from `lm()`	`do_it(n) * { coef(lm(shuffle(y) ~ x))[2] }`	`ptail(obs, null, lower.tail = ...)`

Randomization Confidence Intervals

A **bootstrap confidence interval** estimates the uncertainty of a statistic by repeatedly resampling the observed data *with replacement*. Because it relies on the data itself rather than a mathematical formula, it works for almost any statistic and requires very few distributional assumptions.

Every bootstrap CI in this guide follows the same three steps:

1. **Create a bootstrap distribution** — use `do_it()` to resample with replacement many times, computing the statistic each time.
2. **Calculate the bootstrap SE** — `SE_boot <- sd(boot_dist)`.
3. **Construct the CI** — `obs_stat ± cnorm(C) × SE_boot`.

> **Tip**
>
> **Key SDS1000 functions.** `do_it(n) * { expr }` repeats an expression `n` times. `sample(data, n, replace = TRUE)` resamples with replacement. `get_proportion(v, "cat")` computes a proportion. `resample_pairs(v1, v2)` resamples two paired vectors together. `cnorm(C)` gives the normal critical value for confidence level C.

One Proportion

When to use: A single categorical variable with two outcomes; estimate the population proportion .

Example (class 11): A survey asked 2625 people whether they agreed with "There is only one true love for each person." 1812 disagreed. Compute a 90% bootstrap confidence interval for the proportion who disagreed.

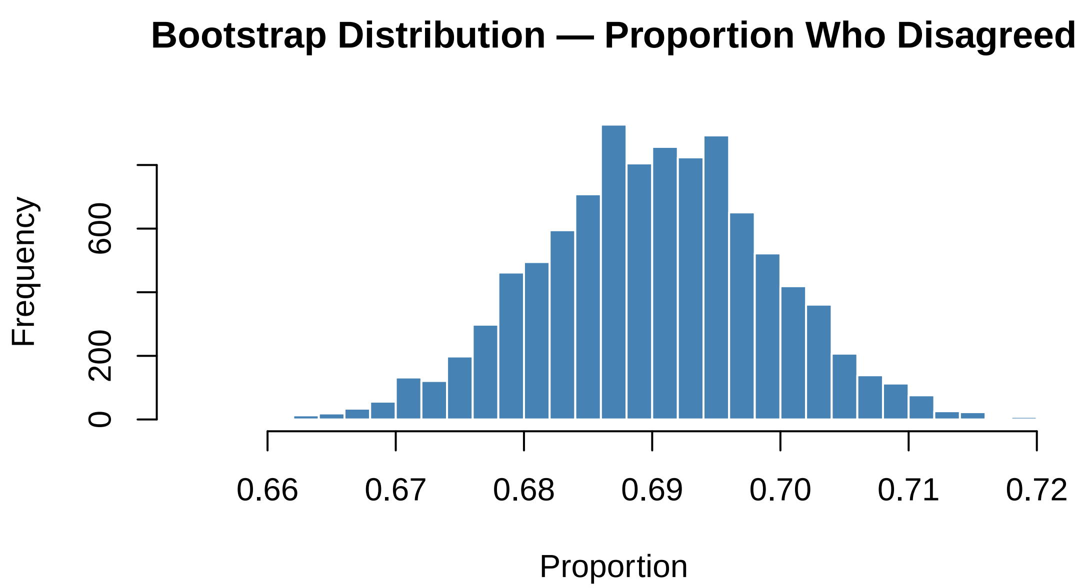
Step 1: Compute the observed statistic

```
believes_in_true_love <- c(rep("FALSE", 1812), rep("TRUE", 2625 - 1812))

obs_prop <- get_proportion(believes_in_true_love, "FALSE")
obs_prop
```

```
    FALSE
0.6902857
```

Step 2: Create the bootstrap distribution

```r
set.seed(2292)

n <- length(believes_in_true_love)

boot_dist <- do_it(10000) * {
  boot_sample <- sample(believes_in_true_love, n, replace = TRUE)
  get_proportion(boot_sample, "FALSE")
}

hist(boot_dist, breaks = 30,
     main = "Bootstrap Distribution - Proportion Who Disagreed",
     xlab = "Proportion",
     col = "steelblue", border = "white")
```



Step 3: Compute the SE and construct the CI

```r
SE_boot <- sd(boot_dist)

CI <- c(obs_prop - cnorm(0.90) * SE_boot,
        obs_prop + cnorm(0.90) * SE_boot)
CI
```

```
    FALSE     FALSE
0.6754056 0.7051658
```

We are 90% confident that the true proportion of people who disagree that there is only one true love for each person is between 0.675 and 0.705.

One Mean

When to use: A single quantitative variable; estimate the population mean .

Example (class 11/12): A sample of 50 body temperatures from Mackowiak et al. (1992). Compute a 95% bootstrap confidence interval for the mean body temperature.

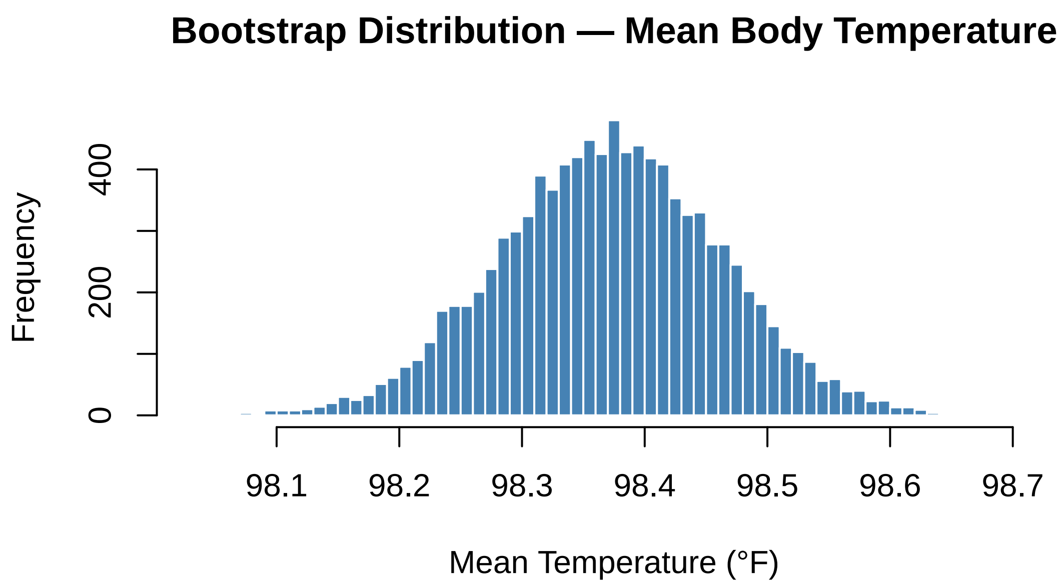
Step 1: Compute the observed statistic

```r
set.seed(9910)
body_temps <- rnorm(50, mean = 98.25, sd = 0.73)

obs_mean <- mean(body_temps)
obs_mean
```

```
[1] 98.37283
```

Step 2: Create the bootstrap distribution

```r
set.seed(4471)

n <- length(body_temps)

boot_dist <- do_it(10000) * {
  mean(sample(body_temps, n, replace = TRUE))
}

hist(boot_dist, breaks = 80,
     main = "Bootstrap Distribution - Mean Body Temperature",
     xlab = "Mean Temperature (°F)",
     col = "steelblue", border = "white")
```



Step 3: Compute the SE and construct the CI

```
SE_boot <- sd(boot_dist)

CI_95 <- c(obs_mean - cnorm(0.95) * SE_boot,
           obs_mean + cnorm(0.95) * SE_boot)
CI_80 <- c(obs_mean - cnorm(0.80) * SE_boot,
           obs_mean + cnorm(0.80) * SE_boot)
CI_99 <- c(obs_mean - cnorm(0.99) * SE_boot,
           obs_mean + cnorm(0.99) * SE_boot)

CI_80; CI_95; CI_99
```

```
[1] 98.25793 98.48773
```

```
[1] 98.19711 98.54855
```

```
[1] 98.14189 98.60376
```

We are 95% confident that the mean human body temperature is between 98.2°F and 98.55°F.

> **Tip**
>
> **Wider CI = more confident.** A 99% CI is wider than a 95% CI, which is wider than an 80% CI. There is always a trade-off: greater confidence requires sacrificing precision.

Difference of Two Means

When to use: A quantitative variable measured in two independent groups; estimate $\mu_1 - \mu_2$. Resample each group separately with replacement, keeping group sizes fixed.

Example: Using the calcium supplement study from class 17, compute a 95% bootstrap confidence interval for the difference in mean blood pressure decrease (treatment − control).

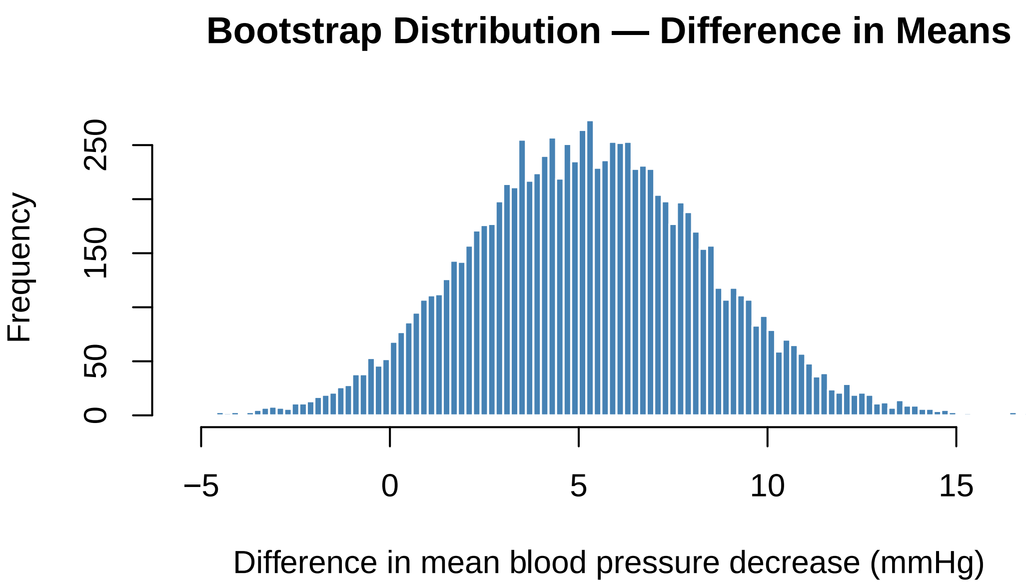
Step 1: Compute the observed statistic

```
treat   <- c( 7, -4, 18, 17, -3, -5,  1, 10, 11, -2)
control <- c(-1, 12, -1, -3,  3, -5,  5,  2, -11, -1, -3)

obs_diff <- mean(treat) - mean(control)
obs_diff
```

```
[1] 5.272727
```

Step 2: Create the bootstrap distribution

```
set.seed(6174)

n_treat   <- length(treat)
n_control <- length(control)

boot_dist <- do_it(10000) * {
  boot_treat   <- sample(treat,   n_treat,   replace = TRUE)
  boot_control <- sample(control, n_control, replace = TRUE)
  mean(boot_treat) - mean(boot_control)
}

hist(boot_dist, breaks = 80,
     main = "Bootstrap Distribution - Difference in Means",
     xlab = "Difference in mean blood pressure decrease (mmHg)",
     col = "steelblue", border = "white")
```



Step 3: Compute the SE and construct the CI

```
SE_boot <- sd(boot_dist)

CI <- c(obs_diff - cnorm(0.95) * SE_boot,
        obs_diff + cnorm(0.95) * SE_boot)
CI
```

```
[1] -0.8689757 11.4144302
```

The 95% CI includes zero, consistent with the hypothesis test result. However, the interval also extends well above zero, meaning a meaningful calcium effect remains plausible.

Correlation

When to use: Two paired quantitative variables; estimate the population correlation . Because x and y are paired, use `resample_pairs()` so that every resampled (x, y) pair stays matched.

Example (class 18): Sugar content and calorie content for a sample of 77 breakfast cereals. Compute a 95% bootstrap confidence interval for the correlation.

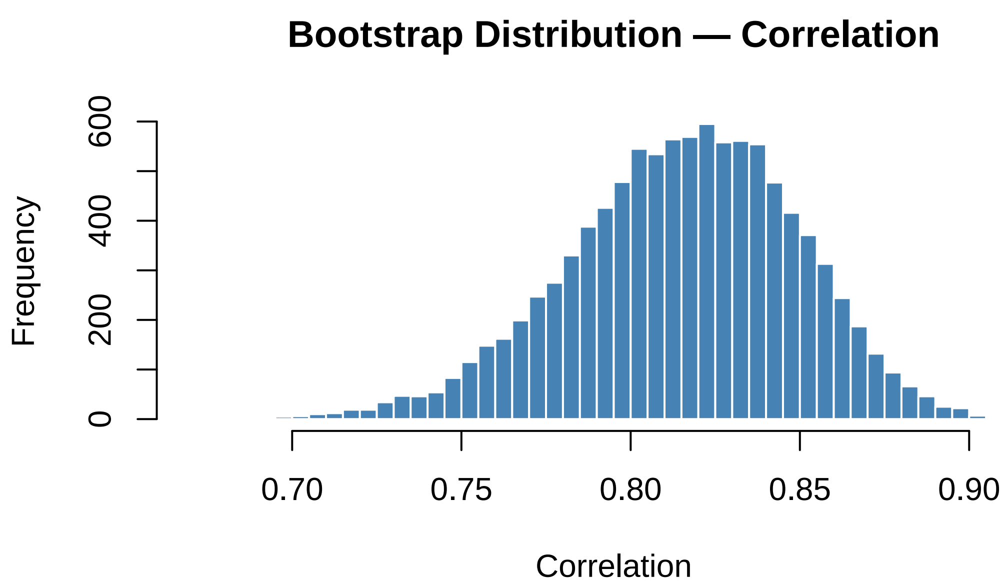
Step 1: Compute the observed statistic

```r
set.seed(1123)
n        <- 77
sugar    <- runif(n, 0, 15)
calories <- 90 + 3.5 * sugar + rnorm(n, sd = 12)

obs_cor <- cor(sugar, calories)
obs_cor
```

```
[1] 0.8149169
```

Step 2: Create the bootstrap distribution

```r
set.seed(8833)

boot_dist <- do_it(10000) * {
  resampled          <- resample_pairs(sugar, calories)
  cor(resampled$vector1, resampled$vector2)
}

hist(boot_dist, breaks = 80,
     main = "Bootstrap Distribution - Correlation",
     xlab = "Correlation",
     col = "steelblue", border = "white")
```



Step 3: Compute the SE and construct the CI

```r
SE_boot <- sd(boot_dist)

CI <- c(obs_cor - cnorm(0.95) * SE_boot,
        obs_cor + cnorm(0.95) * SE_boot)
CI
```

```
[1] 0.7485002 0.8813337
```

We are 95% confident that the true correlation between sugar and calorie content in cereals is between 0.749 and 0.881.

> **Tip**
>
> **Why `resample_pairs()` and not `sample(..., replace = TRUE)`?** For paired data, we must resample observations together to preserve the relationship between x and y. Resampling x and y independently would destroy the very correlation we are trying to measure.

Regression Slope

When to use: Two paired quantitative variables; estimate the slope of a simple linear regression model. Use `resample_pairs()` to keep pairs intact.

Example (class 25): Cigarette consumption per capita and lung cancer rates across 44 U.S. states. Compute a 95% bootstrap confidence interval for the slope.

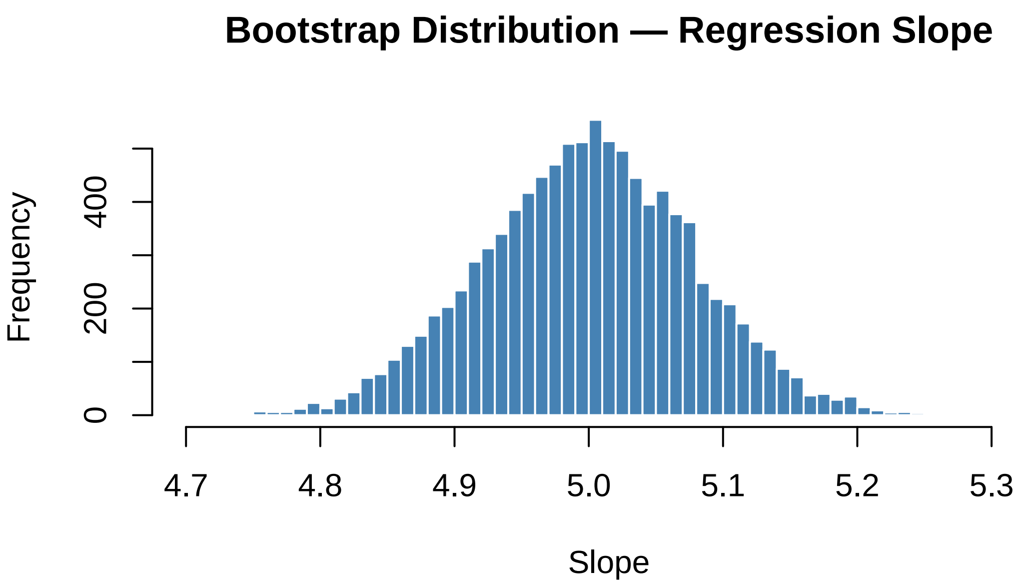
Step 1: Compute the observed statistic

```r
set.seed(2947)
cigs_per_capita <- runif(44, min = 10, max = 45)
cancer_rate     <- 2 + 0.005 * cigs_per_capita * 1000 + rnorm(44, sd = 5)

lung_lm   <- lm(cancer_rate ~ cigs_per_capita)
obs_slope <- coef(lung_lm)["cigs_per_capita"]
obs_slope
```

```
cigs_per_capita
       4.997922
```

Step 2: Create the bootstrap distribution

```r
set.seed(3109)

boot_dist <- do_it(10000) * {
  resampled        <- resample_pairs(cigs_per_capita, cancer_rate)
  resampled_cigs   <- resampled$vector1
  resampled_cancer <- resampled$vector2
  coef(lm(resampled_cancer ~ resampled_cigs))["resampled_cigs"]
}

hist(boot_dist, breaks = 80,
     main = "Bootstrap Distribution - Regression Slope",
     xlab = "Slope",
     col = "steelblue", border = "white")
```



Step 3: Compute the SE and construct the CI

```r
SE_boot <- sd(boot_dist)

CI <- c(obs_slope - cnorm(0.95) * SE_boot,
        obs_slope + cnorm(0.95) * SE_boot)
CI
```

```
cigs_per_capita cigs_per_capita
       4.844075        5.151769
```

We are 95% confident that for each additional hundred cigarettes smoked per capita, the lung cancer rate increases by between 4.8441 and 5.1518 cases per 100,000 people.

Quick Reference

Parameter	Resample method	CI formula
One proportion	`sample(data, n, replace=TRUE)` → `get_proportion()`	`obs ± cnorm(C) * sd(boot_dist)`
One mean	`sample(data, n, replace=TRUE)` → `mean()`	`obs ± cnorm(C) * sd(boot_dist)`
Diff. of two means	`sample(group, n, replace=TRUE)` for each group	`obs ± cnorm(C) * sd(boot_dist)`
Correlation	`resample_pairs(x, y)` → `cor()`	`obs ± cnorm(C) * sd(boot_dist)`
Regression slope	`resample_pairs(x, y)` → `coef(lm(...))`	`obs ± cnorm(C) * sd(boot_dist)`

Parametric Hypothesis Tests

This guide covers every parametric hypothesis test from S&DS 1000. Each test follows the same five steps used in class:

1. **State** the null and alternative hypotheses
2. **Calculate** the observed test statistic (and check conditions)
3. **Visualize** the null distribution and mark the observed statistic
4. **Calculate** the p-value
5. **Make a decision**

> ⚠ Warning
>
> **A note on assumptions.** Parametric tests rely on mathematical approximations that are only valid when certain conditions are met (e.g., large enough sample sizes, approximate normality). Always check conditions before interpreting a p-value. The randomization methods above are more robust when conditions are uncertain.

One Proportion

When to use: One categorical variable with two outcomes; test whether the population proportion equals a specific value .

Conditions: $n\pi_0 \geq 10$ and $n(1-\pi_0) \geq 10$

Test statistic:

$$z = \frac{\hat{p} - \pi_0}{SE_0}, \qquad SE_0 = \sqrt{\frac{\pi_0(1-\pi_0)}{n}}$$

Example (class 20): From 1982–1994 there were 128 penalty shots in the World Cup. Goalkeepers correctly guessed the kick direction 41% of the time. Test whether goalies guessed correctly **less than** 50% of the time.

Step 1: State the hypotheses

$$H_0 : \pi = 0.50 \qquad H_A : \pi < 0.50$$

Steps 2–4: Calculate, visualize, p-value

```
pi_0  <- 0.50
n     <- 128
p_hat <- 0.41
```

```r
# Check conditions
n * pi_0; n * (1 - pi_0)
```

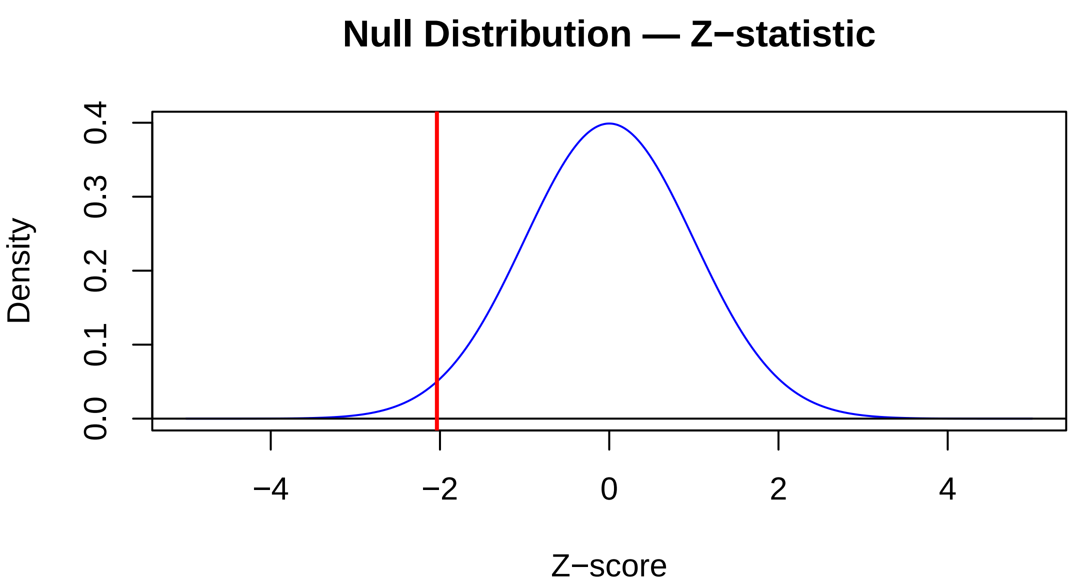
```
[1] 64
```

```
[1] 64
```

```r
SE     <- sqrt(pi_0 * (1 - pi_0) / n)
z_stat <- (p_hat - pi_0) / SE
z_stat
```

```
[1] -2.036468
```

```r
plot_norm(col = "blue",
          main = "Null Distribution - Z-statistic",
          xlab = "Z-score", ylab = "Density")
abline(v = z_stat, col = "red", lwd = 2)
```




```r
p_value <- pnorm(z_stat, mean = 0, sd = 1, lower.tail = TRUE)
p_value
```

```
[1] 0.02085172
```

Step 5: Make a decision

Since p $= 0.0209 < 0.05$, we **reject** H_0 and conclude that goalkeepers guess the correct direction less than 50% of the time.

Two Proportions

When to use: A binary outcome measured in two independent groups; test whether the two population proportions are equal.

Conditions: At least 10 successes and 10 failures in each group.

Test statistic (using the unpooled standard error):

$$z = \frac{\hat{p}_1 - \hat{p}_2}{SE}, \qquad SE = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$$

Example: In a clinical trial, 245 patients received a new vaccine and 255 received a placebo. Of the vaccinated group 32 got the flu; of the placebo group 56 got the flu. Test whether the vaccine **reduces** the proportion who get the flu.

Step 1: State the hypotheses

$$H_0: \pi_1 = \pi_2 \qquad H_A: \pi_1 < \pi_2$$

Steps 2–4: Calculate, visualize, p-value

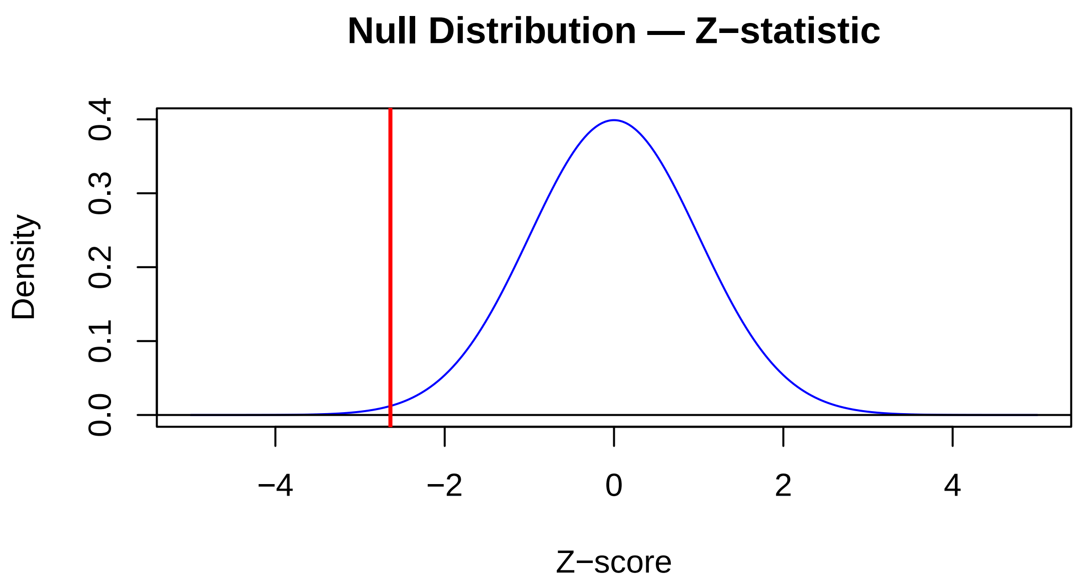
```r
x1 <- 32;  n1 <- 245   # vaccinated
x2 <- 56;  n2 <- 255   # placebo

p_hat1 <- x1 / n1
p_hat2 <- x2 / n2

# Check conditions
c(x1, n1 - x1, x2, n2 - x2)
```

```
[1]  32 213  56 199
```

```r
SE     <- sqrt((p_hat1*(1-p_hat1)/n1) + (p_hat2*(1-p_hat2)/n2))
z_stat <- (p_hat1 - p_hat2) / SE
z_stat
```

```
[1] -2.64097
```

```r
plot_norm(col = "blue",
          main = "Null Distribution - Z-statistic",
          xlab = "Z-score", ylab = "Density")
abline(v = z_stat, col = "red", lwd = 2)
```




```
p_value <- pnorm(z_stat, lower.tail = TRUE)
p_value
```

```
[1] 0.004133447
```

Step 5: Make a decision

Since p = 0.0041 $<$ 0.05, we **reject** H_0 and conclude that the vaccine significantly reduces the proportion of flu cases.

More Than Two Proportions

When to use: One categorical variable with three or more outcomes; test whether the observed counts match a set of expected proportions.

Conditions: Expected count in each cell ≥ 5

Test statistic:

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}, \qquad df = k - 1$$

Example (class 23/24): A Yale professor gathered birth months for 198 randomly selected Yale students. Are students equally likely to be born in any month?

Step 1: State the hypotheses

$$H_0 : \pi_{Jan} = \pi_{Feb} = \cdots = \pi_{Dec} = \tfrac{1}{12}$$
$$H_A : \text{at least one monthly proportion differs from } \tfrac{1}{12}$$

Steps 2–4: Calculate, visualize, p-value

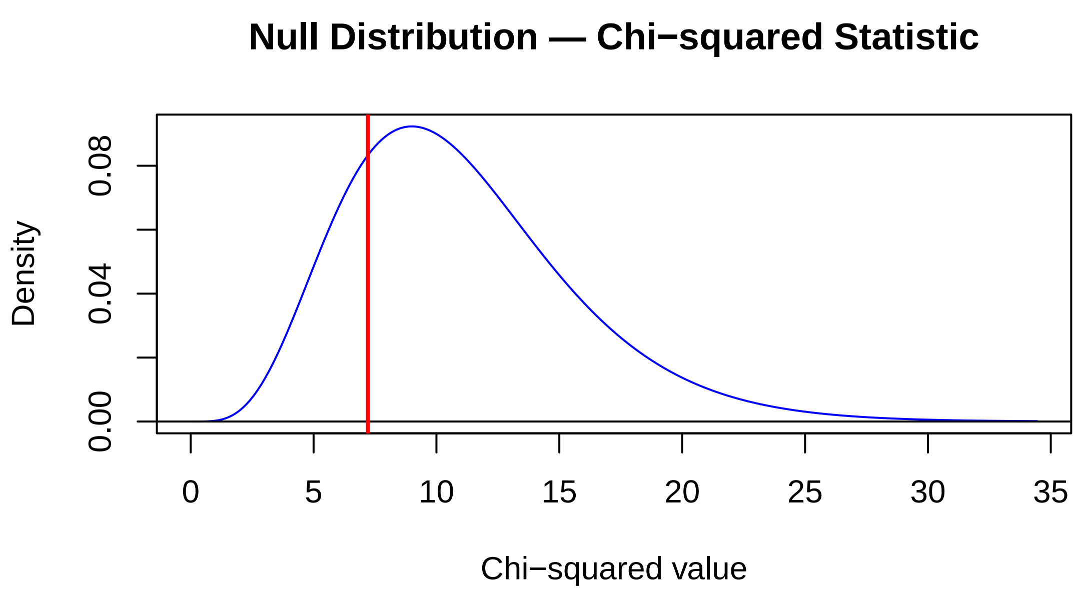
```r
observed_counts <- c(14, 11, 21, 17, 15, 13, 19, 16, 18, 22, 14, 18)
names(observed_counts) <- month.abb
expected_counts <- rep(sum(observed_counts) / 12, 12)

expected_counts[1]   # all ~16.5 - conditions met
```

```
[1] 16.5
```

```r
chi_sq_stat <- sum((observed_counts - expected_counts)^2 / expected_counts)
chi_sq_stat
```

```
[1] 7.212121
```

```r
df <- length(observed_counts) - 1

plot_chisq(df = df, col = "blue",
           main = "Null Distribution - Chi-squared Statistic",
           xlab = "Chi-squared value", ylab = "Density")
abline(v = chi_sq_stat, col = "red", lwd = 2)
```




```r
p_value <- pchisq(chi_sq_stat, df = df, lower.tail = FALSE)
p_value
```

```
[1] 0.7816545
```

Step 5: Make a decision

Since p = 0.782 > 0.05, we **fail to reject** H_0. There is not sufficient evidence that Yale students' birth months deviate from a uniform distribution.

One Mean

When to use: One quantitative variable; test whether the population mean equals a specific value .

Conditions: $n \geq 30$, or the population is approximately normal.

Test statistic:

$$t = \frac{\bar{x} - \mu_0}{SE}, \qquad SE = \frac{s}{\sqrt{n}}, \qquad df = n - 1$$

Example (class 21): Air Force cadets tested 42 bags of Chips Ahoy! cookies and found a mean of 1261.6 chips, sd = 117.6. Test whether the mean exceeds 1000.

Step 1: State the hypotheses

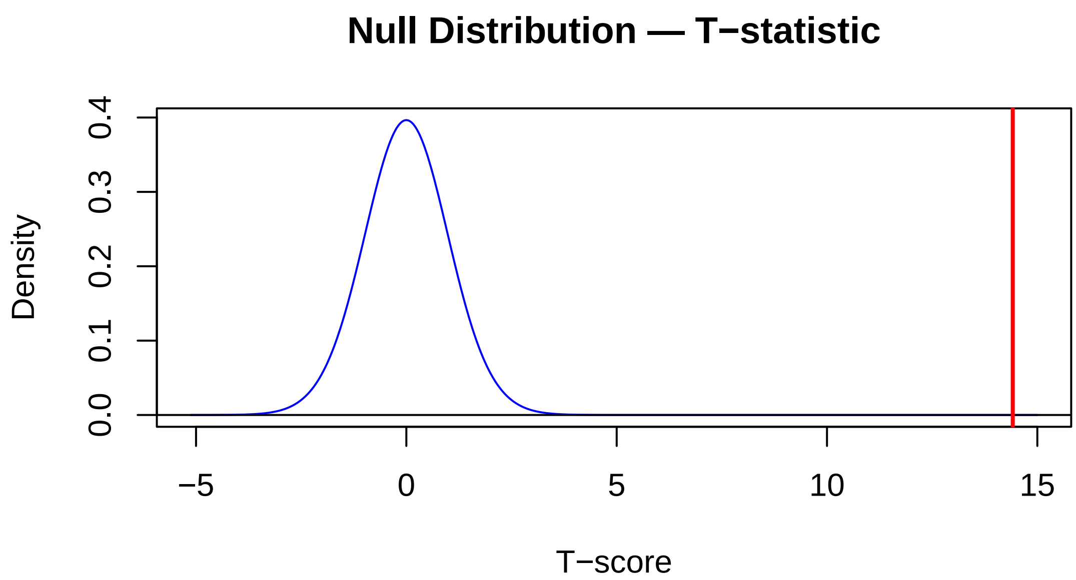
$$H_0 : \mu = 1000 \qquad H_A : \mu > 1000$$

Steps 2–4: Calculate, visualize, p-value

```r
n     <- 42
x_bar <- 1261.6
s     <- 117.6
mu_0  <- 1000

SE     <- s / sqrt(n)
t_stat <- (x_bar - mu_0) / SE
t_stat
```

```
[1] 14.41634
```

```r
df <- n - 1
plot_t(df, to = 15, col = "blue",
       main = "Null Distribution - T-statistic",
       xlab = "T-score", ylab = "Density")
abline(v = t_stat, col = "red", lwd = 2)
```




```
p_value <- pt(t_stat, df = df, lower.tail = FALSE)
p_value
```

```
[1] 5.956987e-18
```

Step 5: Make a decision

Since p 0, we **reject** H_0 and conclude the average chip count exceeds 1000.

Two Means: Independent Samples

When to use: A quantitative variable measured in two independent groups; test whether the group means are equal.

Conditions: Each group has $n \geq 30$, or both populations are approximately normal.

Test statistic (Welch's t-test):

$$t = \frac{(\bar{x}_1 - \bar{x}_2)}{SE_{diff}}, \qquad SE_{diff} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \qquad df = \min(n_1 - 1,\ n_2 - 1)$$

Example (class 22): A study of 2295 American men found that right-handers (n = 2027) earned \$13.10/hr and left-handers (n = 268) earned \$13.40/hr, both with sd = \$7.90. Test whether there is a difference in earnings.

Step 1: State the hypotheses

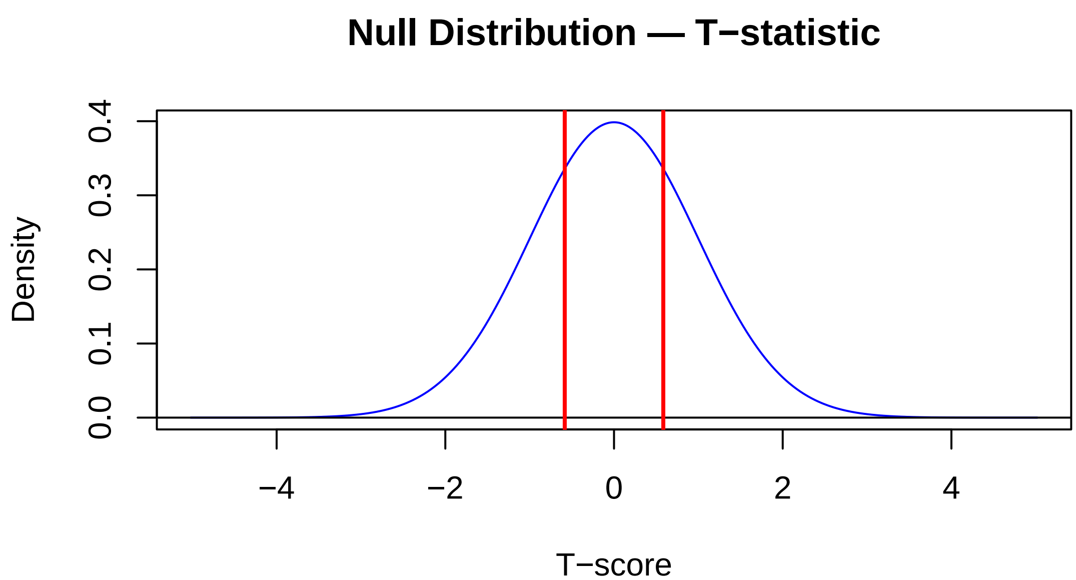
$$H_0 : \mu_1 = \mu_2 \qquad H_A : \mu_1 \neq \mu_2$$

Steps 2–4: Calculate, visualize, p-value

```
n1 <- 2027;  x_bar1 <- 13.10;  s1 <- 7.90
n2 <- 268;   x_bar2 <- 13.40;  s2 <- 7.90

SE_diff <- sqrt((s1^2/n1) + (s2^2/n2))
t_stat  <- (x_bar1 - x_bar2) / SE_diff
t_stat
```

```
[1] -0.5842478
```

```
df <- min(n1 - 1, n2 - 1)
plot_t(df, col = "blue",
       main = "Null Distribution - T-statistic",
       xlab = "T-score", ylab = "Density")
abline(v =  t_stat, col = "red", lwd = 2)
abline(v = -t_stat, col = "red", lwd = 2)
```




```
p_value <- 2 * pt(-abs(t_stat), df = df)
p_value
```

```
[1] 0.5595469
```

Step 5: Make a decision

Since p $= 0.56 > 0.05$, we **fail to reject** H_0. There is not sufficient evidence of a difference in hourly earnings between right- and left-handed men.

Two Means: Paired Samples

When to use: Observations come in natural pairs (e.g., before/after on the same subject). Compute within-pair differences and run a one-sample t-test.

Test statistic:

$$t = \frac{\bar{d}}{SE_d}, \qquad SE_d = \frac{s_d}{\sqrt{n}}, \qquad df = n - 1$$

Example (class 22): Exam scores for 10 students on Exam 1 and Exam 2. Test whether students scored higher on Exam 2 on average.

Step 1: State the hypotheses

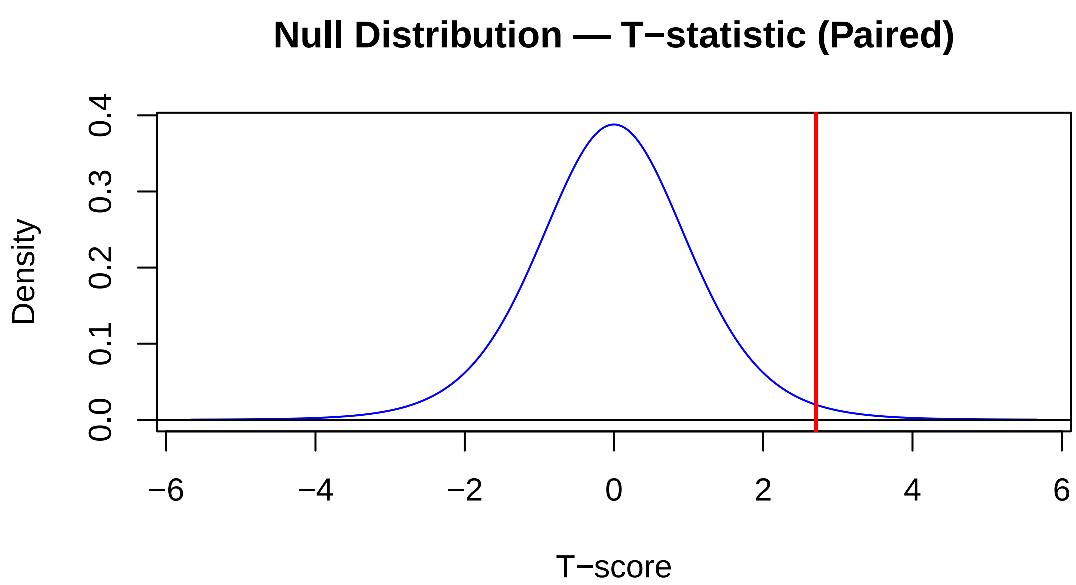
$$H_0 : \mu_d = 0 \qquad H_A : \mu_d > 0$$

Steps 2–4: Calculate, visualize, p-value

```
exam1 <- c(72, 95, 56, 87, 80, 98, 74, 85, 77, 62)
exam2 <- c(78, 96, 72, 89, 80, 95, 86, 87, 82, 72)

d      <- exam2 - exam1
n      <- length(d)
SE_d   <- sd(d) / sqrt(n)
t_stat <- mean(d) / SE_d
t_stat
```

```
[1] 2.709344
```

```
df <- n - 1
plot_t(df, col = "blue",
       main = "Null Distribution - T-statistic (Paired)",
       xlab = "T-score", ylab = "Density")
abline(v = t_stat, col = "red", lwd = 2)
```




```
p_value <- pt(t_stat, df = df, lower.tail = FALSE)
p_value
```

```
[1] 0.01201164
```

Step 5: Make a decision

Since p $= 0.012 < 0.05$, we **reject** H_0 and conclude that students scored significantly higher on Exam 2.

More Than Two Means: ANOVA

When to use: A quantitative variable measured across three or more independent groups; test whether all group means are equal.

Conditions: Approximately normal residuals; roughly equal variance across groups; independent observations.

Test statistic:

$$F = \frac{MSG}{MSE}, \qquad df_1 = k - 1, \quad df_2 = N - k$$

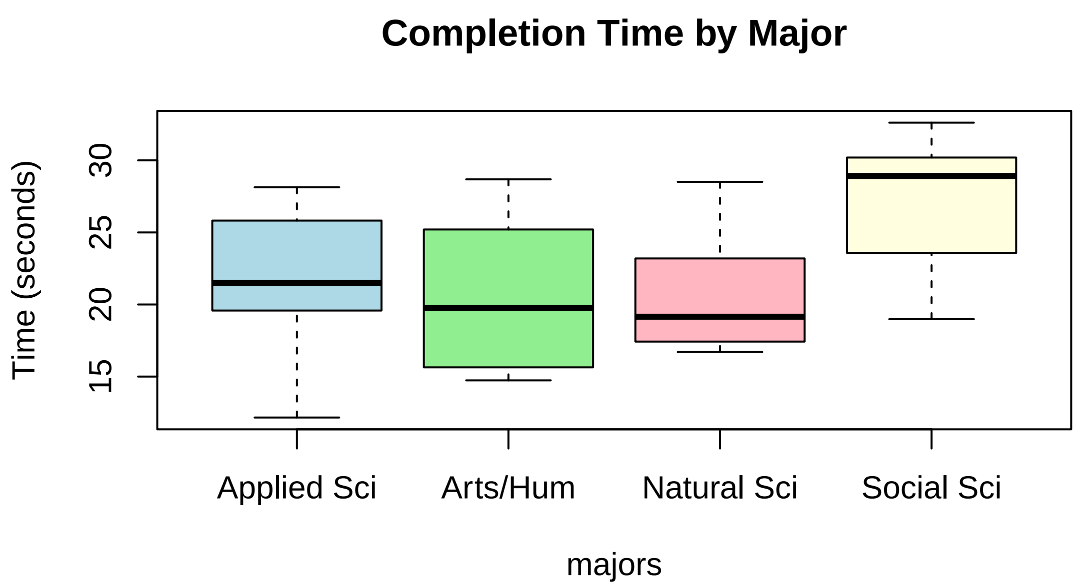
Example (class 24): Hope College students were timed completing a Sudoku-like puzzle. Majors were grouped into four categories. Is mean completion time the same across majors?

Step 1: State the hypotheses

$$H_0 : \mu_{as} = \mu_{ns} = \mu_{ss} = \mu_{ah} \qquad H_A : \text{at least one mean differs}$$

Steps 2–4: Calculate, visualize, p-value

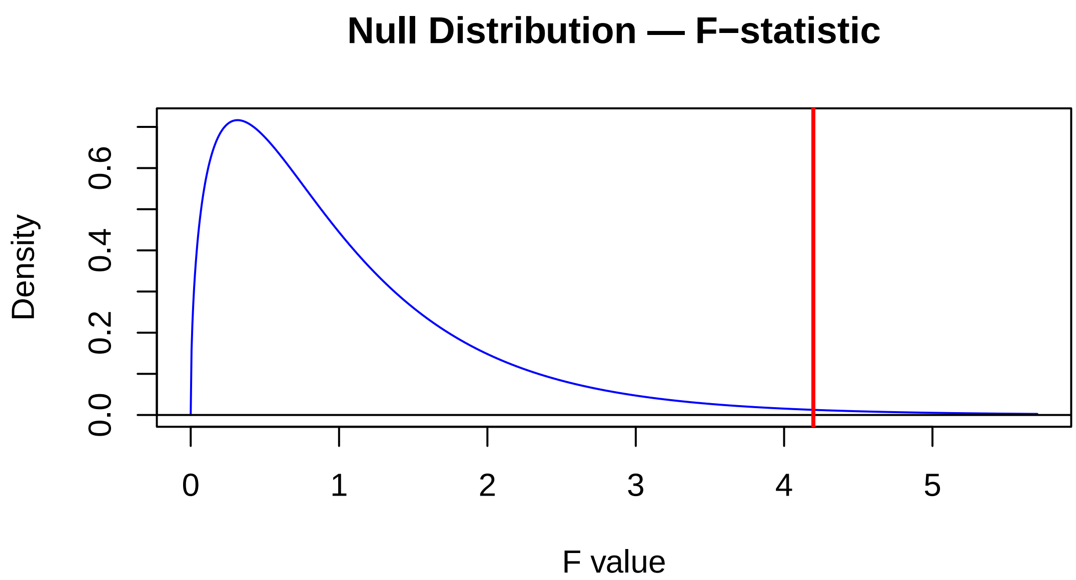
```r
set.seed(4455)
completion_times <- c(rnorm(10, mean = 22, sd = 5),
                      rnorm(10, mean = 20, sd = 5),
                      rnorm(10, mean = 26, sd = 5),
                      rnorm(10, mean = 21, sd = 5))
majors <- factor(rep(c("Applied Sci", "Natural Sci", "Social Sci", "Arts/Hum"),
                     each = 10))

boxplot(completion_times ~ majors,
        main = "Completion Time by Major",
        ylab = "Time (seconds)",
        col  = c("lightblue", "lightgreen", "lightpink", "lightyellow"))
```




```r
f_stat <- get_F_stat(completion_times, majors)
df1 <- length(unique(majors)) - 1
df2 <- length(completion_times) - length(unique(majors))
f_stat
```

```
[1] 4.196999
```

```r
plot_f(df1, df2, col = "blue",
       main = "Null Distribution - F-statistic",
       xlab = "F value", ylab = "Density")
abline(v = f_stat, col = "red", lwd = 2)
```




```r
p_value <- pf(f_stat, df1, df2, lower.tail = FALSE)
p_value
```

```
[1] 0.01204589
```

Step 5: Make a decision

p = 0.012: we reject H — at least one group mean differs.

Correlation

When to use: Two quantitative variables; test whether the linear association between them is significantly different from zero.

Conditions: Both variables approximately normal, or n is large; the relationship is linear.

Test statistic:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \qquad df = n-2$$

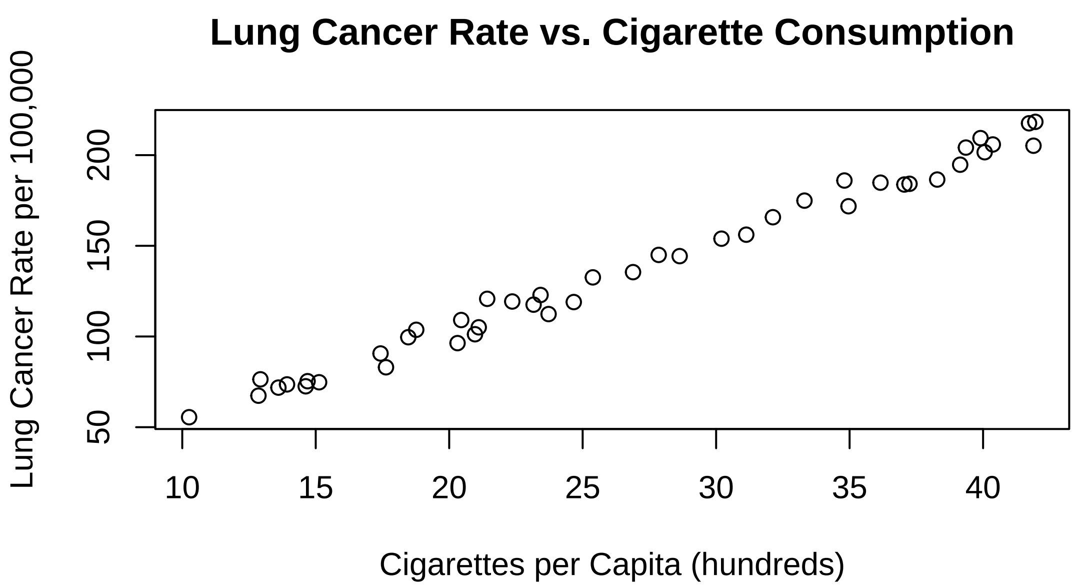
Example (class 25): Data on cigarette consumption per capita and lung cancer rates across 44 U.S. states. Is there evidence of a positive linear association?

Step 1: State the hypotheses

$$H_0 : \rho = 0 \qquad H_A : \rho > 0$$

Steps 2–4: Calculate, visualize, p-value

```r
set.seed(2947)
cigs_per_capita <- runif(44, min = 10, max = 45)
cancer_rate     <- 2 + 0.005 * cigs_per_capita * 1000 + rnorm(44, sd = 5)

plot(cigs_per_capita, cancer_rate,
     xlab = "Cigarettes per Capita (hundreds)",
     ylab = "Lung Cancer Rate per 100,000",
     main = "Lung Cancer Rate vs. Cigarette Consumption")
```



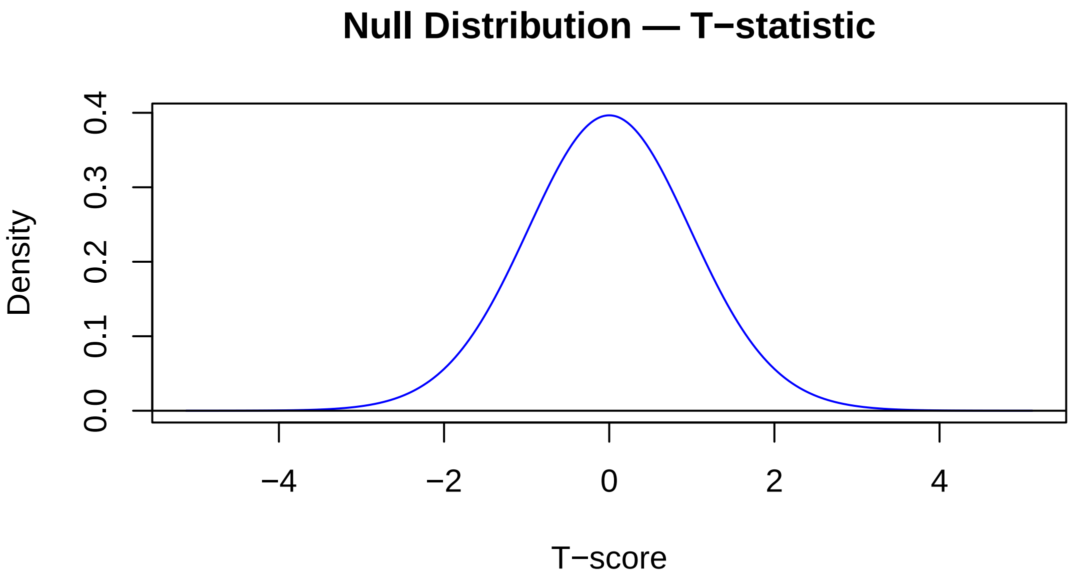

```r
n <- length(cigs_per_capita)
r <- cor(cigs_per_capita, cancer_rate)
r
```

```
[1] 0.9945126
```

```r
t_stat <- r * sqrt(n - 2) / sqrt(1 - r^2)
t_stat
```

```
[1] 61.60758
```

```r
df <- n - 2
plot_t(df, col = "blue",
       main = "Null Distribution - T-statistic",
       xlab = "T-score", ylab = "Density")
abline(v = t_stat, col = "red", lwd = 2)
```




```
p_value <- pt(t_stat, df = df, lower.tail = FALSE)
p_value
```

```
[1] 4.094627e-43
```

Step 5: Make a decision

Since p = 0, we **reject** H_0 and conclude that there is a significant positive linear association between cigarette consumption and lung cancer rate.

> **Tip**
>
> **Shortcut:** `cor.test(x, y, alternative = "greater")` performs all of steps 2–4 in one call and returns the same t-statistic, df, and p-value.

Simple Linear Regression

When to use: Two quantitative variables where you want to model y as a linear function of x **and** test whether the slope differs from zero. Testing H : = 0 is equivalent to testing H : = 0 — they yield identical t-statistics and p-values.

Conditions: Linear relationship; approximately normal residuals; constant variance; independent observations.

Test statistic for the slope:

$$t = \frac{\hat{\beta}_1}{SE_{\hat{\beta}_1}}, \qquad df = n - 2$$

Example (class 25): Using the same cigarette/lung cancer data, test whether the slope is significantly positive.

Step 1: State the hypotheses

$$H_0 : \beta_1 = 0 \qquad H_A : \beta_1 > 0$$

Steps 2–4: Fit the model, visualize, p-value

```r
lung_lm <- lm(cancer_rate ~ cigs_per_capita)
summary(lung_lm)
```

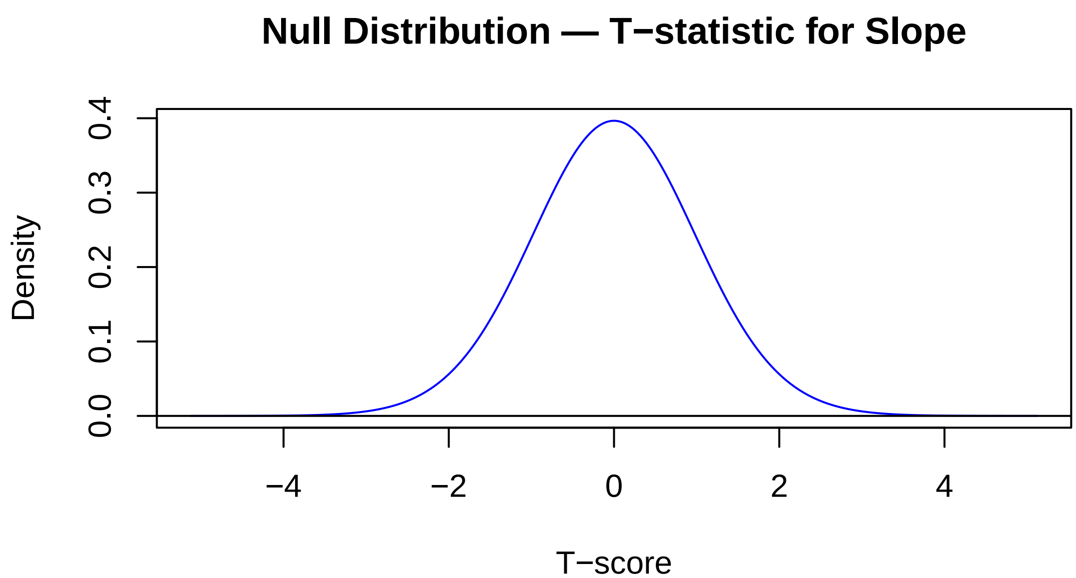
```

Call:
lm(formula = cancer_rate ~ cigs_per_capita)

Residuals:
    Min      1Q  Median      3Q     Max 
-9.1802 -3.8251  0.3541  4.0038 10.7198 

Coefficients:
                Estimate Std. Error t value Pr(>|t|)    
(Intercept)      2.97304    2.31498   1.284    0.206    
cigs_per_capita  4.99792    0.08113  61.608   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.237 on 42 degrees of freedom
Multiple R-squared:  0.9891,	Adjusted R-squared:  0.9888 
F-statistic:  3795 on 1 and 42 DF,  p-value: < 2.2e-16
```

Reading the `Coefficients` table:

Column	Meaning
`Estimate`	Fitted intercept (ˆ) and slope (ˆ)
`Std. Error`	Standard error of each estimate
`t value`	t-statistic testing H : coefficient = 0
`Pr(>|t|)`	Two-sided p-value

```r
fit_sum <- summary(lung_lm)
t_stat  <- fit_sum$coefficients["cigs_per_capita", "t value"]
df      <- lung_lm$df.residual

plot_t(df, col = "blue",
       main = "Null Distribution - T-statistic for Slope",
```

```
       xlab = "T-score", ylab = "Density")
abline(v = t_stat, col = "red", lwd = 2)
```




```
p_two_sided <- fit_sum$coefficients["cigs_per_capita", "Pr(>|t|)"]
p_value     <- p_two_sided / 2
p_value
```

```
[1] 4.094627e-43
```

Step 5: Make a decision

Since p = 0 < 0.05, we **reject** H_0 and conclude that there is a significant positive relationship between cigarette consumption and lung cancer rate.

Confidence interval for the slope

```
confint(lung_lm, level = 0.90)
```

```
                       5 %      95 %
(Intercept)     -0.9206509 6.866724
cigs_per_capita  4.8614732 5.134370
```

Quick Reference

Test	H	Statistic	df	p-value function
One proportion	$\pi = \pi_0$	$z = \frac{\hat{p}-\pi_0}{SE_0}$	—	`pnorm(z, lower.tail=...)`
Two proportions	$\pi_1 = \pi_2$	$z = \frac{\hat{p}_1-\hat{p}_2}{SE}$	—	`pnorm(z, lower.tail=...)`
Multiple proportions	$\pi_i = \pi_0$	$\chi^2 = \sum(O-E)^2/E$	$k-1$	`pchisq(x, df, lower.tail=FALSE)`
One mean	$\mu = \mu_0$	$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$	$n-1$	`pt(t, df, lower.tail=...)`
Two means (indep.)	$\mu_1 = \mu_2$	$t = \frac{\bar{x}_1-\bar{x}_2}{SE_{diff}}$	$\min(n_1, n_2) - 1$	`pt(t, df, lower.tail=...)`
Two means (paired)	$\mu_d = 0$	$t = \frac{\bar{d}}{s_d/\sqrt{n}}$	$n-1$	`pt(t, df, lower.tail=...)`
Multiple means	$\mu_1 = \cdots = \mu_k$	$F = MSG/MSE$	$k-1$, $N-k$	`pf(f, df1, df2, lower.tail=FALSE)`
Correlation	$\rho = 0$	$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$	$n-2$	`pt(t, df, lower.tail=...)`
Regression slope	$\beta_1 = 0$	$t = \hat{\beta}_1/SE$	$n-2$	`summary(lm(...))`

Parametric Confidence Intervals

A **confidence interval** gives a range of plausible values for an unknown population parameter. A parametric CI uses a mathematical formula based on the sampling distribution of the estimator. Every CI in this guide takes the form:

$$\text{point estimate} \pm \text{critical value} \times SE$$

The **critical value** comes from a normal or t-distribution and is found using `cnorm(C)` (normal) or `ct(C, df)` (t) for a confidence level C. The **standard error (SE)** measures the typical variability of the estimator.

One Proportion

When to use: A single categorical variable with two outcomes; estimate the population proportion .

Conditions: $n\hat{p} \geq 10$ and $n(1-\hat{p}) \geq 10$

Formula:

$$\hat{p} \pm z^* \cdot \hat{SE}, \qquad \hat{SE} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Example (class 20): A survey asked 2625 people if they agreed that "There is only one true love for each person." 1812 of the respondents disagreed. Compute a 90% confidence interval for the proportion who disagreed.

```
n     <- 2625
x     <- 1812
p_hat <- x / n
p_hat
```

```
[1] 0.6902857
```

```
# Check conditions
n * p_hat; n * (1 - p_hat)
```

```
[1] 1812
```

```
[1] 813
```

```
SE     <- sqrt(p_hat * (1 - p_hat) / n)
z_star <- cnorm(0.90)

CI <- c(p_hat - z_star * SE, p_hat + z_star * SE)
CI
```

```
[1] 0.6754415 0.7051299
```

We are 90% confident that the true proportion who disagree is between 0.675 and 0.705.

One Mean

When to use: A single quantitative variable; estimate the population mean .

Conditions: $n \geq 30$, or the population is approximately normal.

Formula:

$$\bar{x} \pm t^* \cdot SE, \qquad SE = \frac{s}{\sqrt{n}}, \qquad df = n - 1$$

Example (class 21): A study recorded outdoor hunting activity of n = 55 domestic cats. The mean number of kills per week was 2.4 with a standard deviation of 1.51. Compute a 99% confidence interval for the mean.

```
n     <- 55
x_bar <- 2.4
s     <- 1.51

SE     <- s / sqrt(n)
df     <- n - 1
t_star <- ct(0.99, df)
t_star
```

```
[1] 2.669985
```

```
CI <- c(x_bar - t_star * SE, x_bar + t_star * SE)
CI
```

```
[1] 1.856369 2.943631
```

We are 99% confident that the mean number of kills per week is between 1.856 and 2.944.

Tip

When to use t vs z. For a single mean, always use the t-distribution (`ct()`). The t-distribution has heavier tails than the normal, which accounts for the additional uncertainty from estimating with s. For a proportion, use the normal distribution (`cnorm()`).

Difference of Two Means

When to use: A quantitative variable measured in two independent groups; estimate $\mu_1 - \mu_2$.

Conditions: Each group has $n \geq 30$ or both populations are approximately normal.

Formula (Welch's):

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \cdot SE_{diff}, \qquad SE_{diff} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}, \qquad df = \min(n_1 - 1, n_2 - 1)$$

Example (class 22): A study recorded dietary fiber intake. Compute a 95% confidence interval for the difference in mean daily fiber intake (grams) between males and females.

```r
set.seed(7823)
fiber_males   <- round(rnorm(155, mean = 23.5, sd = 14.2))
fiber_females <- round(rnorm(662, mean = 20.1, sd = 11.8))

n1 <- length(fiber_males);    x_bar1 <- mean(fiber_males);    s1 <- sd(fiber_males)
n2 <- length(fiber_females);  x_bar2 <- mean(fiber_females);  s2 <- sd(fiber_females)

paste("Males:   n =", n1, " mean =", round(x_bar1,2), " sd =", round(s1,2))
```

```
[1] "Males:   n = 155  mean = 24.5   sd = 15.08"
```

```r
paste("Females: n =", n2, " mean =", round(x_bar2,2), " sd =", round(s2,2))
```

```
[1] "Females: n = 662  mean = 20.39  sd = 11.47"
```

```r
SE_diff <- sqrt((s1^2/n1) + (s2^2/n2))
df      <- min(n1 - 1, n2 - 1)
t_star  <- ct(0.95, df)

obs_diff <- x_bar1 - x_bar2
CI <- c(obs_diff - t_star * SE_diff, obs_diff + t_star * SE_diff)
CI
```

```
[1] 1.565513 6.664504
```

The 95% CI includes zero, which suggests we do not have strong evidence of a difference in mean fiber intake between males and females.

Correlation

When to use: Two quantitative variables; construct a confidence interval for the population correlation using **Fisher's z-transformation**.

$$z = \tanh^{-1}(r), \quad SE_z = \frac{1}{\sqrt{n-3}}, \quad CI_z = z \pm z^* \cdot SE_z, \quad CI_r = \tanh(CI_z)$$

Example (class 18/25): Sugar content and calorie content for a sample of 77 breakfast cereals. Compute a 95% confidence interval for the correlation.

```r
set.seed(1123)
n        <- 77
sugar    <- runif(n, 0, 15)
calories <- 90 + 3.5 * sugar + rnorm(n, sd = 12)

corr_result <- cor.test(sugar, calories, conf.level = 0.95)
corr_result$estimate
```

```
      cor
0.8149169
```

```r
corr_result$conf.int
```

```
[1] 0.7228814 0.8785408
attr(,"conf.level")
[1] 0.95
```

We are 95% confident that the true correlation is between 0.723 and 0.879.

Manual calculation using Fisher's z-transformation:

```r
r    <- cor(sugar, calories)
z    <- atanh(r)
SE_z <- 1 / sqrt(n - 3)

z_star <- cnorm(0.95)
ci_z   <- c(z - z_star * SE_z, z + z_star * SE_z)
tanh(ci_z)
```

```
[1] 0.7228814 0.8785408
```

Regression Slope

When to use: Two quantitative variables; estimate how much y changes on average for a one-unit increase in x. Use `confint()` on the fitted model, which uses `qt()` internally with $df = n - 2$.

Example (class 25): Cigarette consumption per capita and lung cancer rates across 44 U.S. states. Compute a 90% confidence interval for the slope.

```
set.seed(2947)
cigs_per_capita <- runif(44, min = 10, max = 45)
cancer_rate     <- 2 + 0.005 * cigs_per_capita * 1000 + rnorm(44, sd = 5)

lung_lm <- lm(cancer_rate ~ cigs_per_capita)

slope <- coef(lung_lm)["cigs_per_capita"]
slope
```

```
cigs_per_capita
       4.997922
```

```
confint(lung_lm, level = 0.90)
```

```
                       5 %     95 %
(Intercept)     -0.9206509 6.866724
cigs_per_capita  4.8614732 5.134370
```

We are 90% confident that for each additional hundred cigarettes smoked per capita, the lung cancer rate increases by between 4.8615 and 5.1344 cases per 100,000 people.

> Tip
>
> **Manual calculation:** You can also compute the CI by hand using `summary(lung_lm)`. The SE of the slope is in the `Std. Error` column, and the critical value is `ct(C, df = n-2)`.

Quick Reference

Parameter	Formula	Critical value	Function
One proportion	$\hat{p} \pm z^* \sqrt{\hat{p}(1-\hat{p})/n}$	`cnorm(C)`	`cnorm()`
One mean	$\bar{x} \pm t^* \cdot s/\sqrt{n}$	`ct(C, n-1)`	`ct()`
Difference –	$(\bar{x}_1 - \bar{x}_2) \pm t^* \cdot SE_{diff}$	`ct(C, min(n ,n )-1)`	`ct()`
Correlation	`cor.test(x, y, conf.level=C)$conf.int`	Fisher z-transform	—
Regression slope	`confint(lm(...), level=C)`	`ct(C, n-2)`	—